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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Simple Harmonic Motion

JEE Main Physics · Previous Year Questions with Answer Key

194	139	2002 – 2026	13	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer) 140 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The frequency of oscillation of a mass m suspended by a spring is v_1. If the length of the spring is cut to half, the same mass oscillates with frequency v_2. The value of v_2/v_1 is ____ .

(A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\sqrt{3}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

A spring stretches by 2 mm when it is loaded with a mass of 200 g . From equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and maximum energy in the spring are ____ Hz and ____ J, respectively.

(Take $g = 10\ m/s^2$)

(A) $\frac{5\sqrt{50}}{\pi}$ and 8×10^{-3} (B) $\frac{5\sqrt{50}}{\pi}$ and 8

(C) $10\sqrt{50}$ and 2×10^{-3} (D) $\frac{5\sqrt{50}}{\pi}$ and 16×10^{-3}

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec . The time required by it to move from $x = A$ to $x = \frac{A}{\sqrt{2}}$ is ____ sec.

(A) 1/4 (B) 5/4 (C) 5/8 (D) 3/8

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Match List - I with List - II.

List - I		List - II	
A.	$\sin^2 \omega t$	I.	Periodic with time period $T = \frac{\pi}{\omega}$ but not simple harmonic motion (SHM)
B.	$\sin^3(2\omega t)$	II.	Periodic with time period $T = \frac{2\pi}{\omega}$ but Not SHM
C.	$\sin(\omega t) + \cos(\pi\omega t)$	III.	Periodic with time period $T = \frac{\pi}{\omega}$ and SHM
D.	$\cos \omega t + \cos 2\omega t$	IV.	Non-periodic

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II (B) A-II, B-I, C-III, D-IV

(C) A-III, B-II, C-IV, D-I (D) A-II, B-I, C-IV, D-III

Open solution

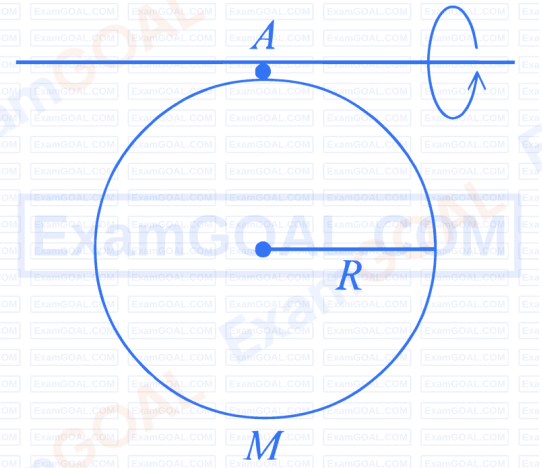
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Q5. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

A uniform disc of radius R and mass M is free to oscillate about the axis A as shown in the figure. For small oscillations the time period is ____ .

(g is acceleration due to gravity)



(A) $2\pi\sqrt{\frac{5R}{4g}}$ (B) $2\pi\sqrt{\frac{2R}{3g}}$ (C) $2\pi\sqrt{\frac{3R}{2g}}$ (D) $2\pi\sqrt{\frac{3R}{g}}$

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

The equation of motion of a particle is given by $x = a\sin(50t + \pi/3)$ cm. The particle will come to rest at time t_1 and it will have zero acceleration at time t_2. The t_1 and t_2 respectively are _____.

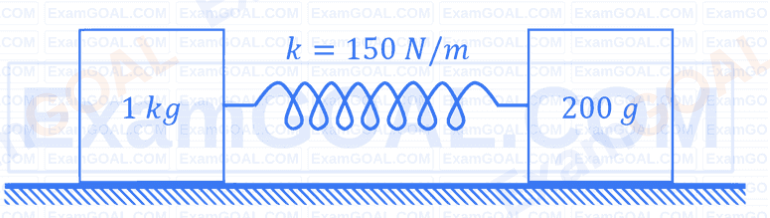
(A) $\frac{\pi}{300}$ s, $\frac{\pi}{75}$ s (B) $\frac{\pi}{75}$ s, $\frac{\pi}{300}$ s

(C) $\frac{\pi}{300}$ s, $\frac{\pi}{25}$ s (D) $\frac{\pi}{50}$ s, $\frac{\pi}{100}$ s

Q7. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

As shown in the figure, a spring is kept in a stretched position with some extension by holding the masses 1 kg and 0.2 kg with a separation more than spring natural length and are released. Assuming the horizontal surface to be frictionless, the angular frequency (in SI unit) of the system is :



(A) 20 (B) 5 (C) 30 (D) 27

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Q8. JEE Main 2026 (Online) 24th January Morning Shift

A cylindrical block of mass M and area of cross section A is floating in a liquid of density ρ and with its axis vertical. When depressed a little and released the block starts oscillating. The period of oscillation is ____

Open solution

(A) $2\pi\sqrt{\frac{\rho A}{Mg}}$ (B) $\pi\sqrt{\frac{2M}{\rho Ag}}$ (C) $2\pi\sqrt{\frac{M}{\rho Ag}}$ (D) $\pi\sqrt{\frac{\rho A}{Mg}}$

Q9. JEE Main 2026 (Online) 24th January Morning Shift

A spring of force constant $15\ \mathrm{N/m}$ is cut into two pieces. If the ratio of their length is $1:3$, then the force constant of smaller piece is ____ $\mathrm{N/m}$.

Open solution

(A) 20 (B) 45 (C) 60 (D) 15

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

A simple pendulum of string length 30 cm performs 20 oscillations in 10 s . The length of the string required for the pendulum to perform 40 oscillations in the same time duration is ____ cm . [Assume that the mass of the pendulum remains same.]

Open solution

(A) 0.75 (B) 7.5 (C) 15 (D) 120

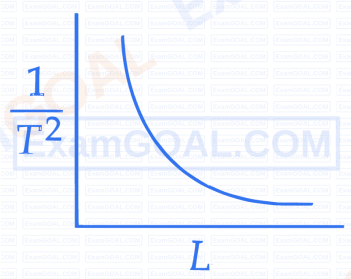
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Q11. JEE Main 2026 (Online) 22nd January Evening Shift

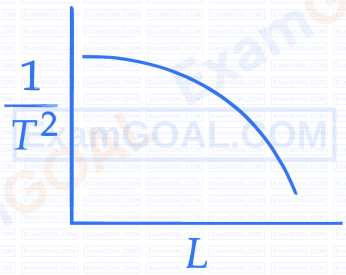
Using a simple pendulum experiment g is determind by measuring its time period T. Which of the following plots represent the correct relation between the pendulum length L and time period T ?

Open solution

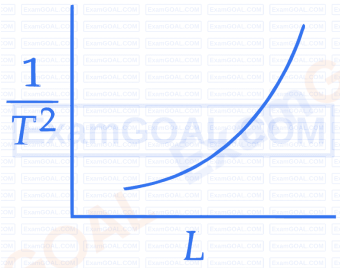
(A)



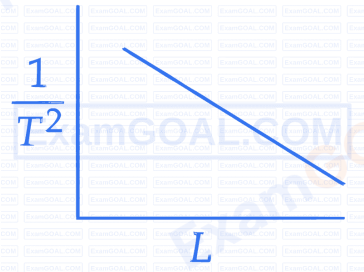
(B)



(C)



(D)



Q12. JEE Main 2026 (Online) 21st January Evening Shift

The kinetic energy of a simple harmonic oscillator is oscillating with angular frequency of 176 rad/s. The frequency of this simple harmonic oscillator is _____ Hz. **[take** $\pi = \frac{22}{7}$ **]**

Open solution

(A) 88 (B) 14 (C) 28 (D) 176

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Q13. JEE Main 2025 (Online) 4th April Morning Shift

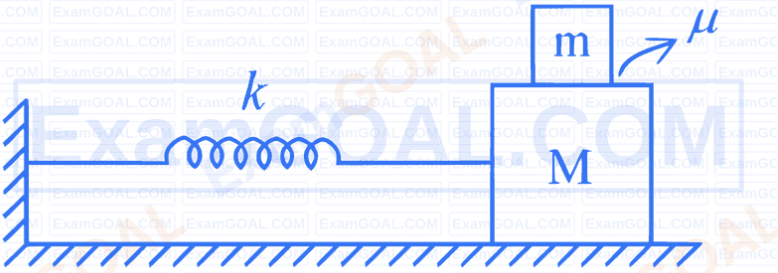
Open solution

Two simple pendulums having lengths l_1 and l_2 with negligible string mass undergo angular displacements θ_1 and θ_2, from their mean positions, respectively. If the angular accelerations of both pendulums are same, then which expression is correct?

(A) $\theta_1 l_2 = \theta_2 l_1$ (B) $\theta_1 l_1 = \theta_2 l_2$ (C) $\theta_1 l_2^2 = \theta_2 l_1^2$ (D) $\theta_1 l_1^2 = \theta_2 l_2^2$

Q14. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution



Two blocks of masses m and $M, (M > m)$, are placed on a frictionless table as shown in figure. A massless spring with spring constant k is attached with the lower block. If the system is slightly displaced and released, then ($\mu =$ coefficient of friction between the two blocks)

A. The time period of small oscillation of the two blocks is $T = 2\pi\sqrt{\frac{(m+M)}{k}}$

B. The acceleration of the blocks is $a = -\frac{kx}{M+m}$ ($x =$ displacement of the blocks from the mean position)

C. The magnitude of the frictional force on the upper block is $\frac{m\mu|x|}{M+m}$

D. The maximum amplitude of the upper block, if it does not slip, is $\frac{\mu(M+m)g}{k}$

E. Maximum frictional force can be $\mu(\mathrm{M}+\mathrm{m})\mathrm{g}$.

Choose the correct answer from the options given below :

(A) B, C, D Only (B) C, D, E Only (C) A, B, D Only (D) A, B, C Only

Q15. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A particle is subjected to two simple harmonic motions as : $x_1 = \sqrt{7}\sin 5t$ cm and $x_2 = 2\sqrt{7}\sin\left(5t + \frac{\pi}{3}\right)$cm where x is displacement and t is time in seconds. The maximum acceleration of the particle is $x \times 10^{-2}$ ms^{-2}. The value of x is :

(A) $5\sqrt{7}$ (B) 125 (C) $25\sqrt{7}$ (D) 175

Q16. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is

(A) $\sqrt{\frac{k_2}{k_1}}$ (B) $\sqrt{\frac{k_1}{k_2}}$ (C) $\frac{k_2}{k_1}$ (D) $\frac{k_1}{k_2}$

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Q17. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Time period of a simple pendulum is longer at the top of a mountain than that at the base of the mountain.

Reason (R) : Time period of a simple pendulum decreases with increasing value of acceleration due to gravity and vice-versa.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

(B) **(A)** is true but **(R)** is false.

(C) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(D) **(A)** is false but **(R)** is true.

Q18. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Knowing initial position x_0 and initial momentum p_0 is enough to determine the position and momentum at any time t for a simple harmonic motion with a given angular frequency ω.

Reason (R) : The amplitude and phase can be expressed in terms of x_0 and p_0.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

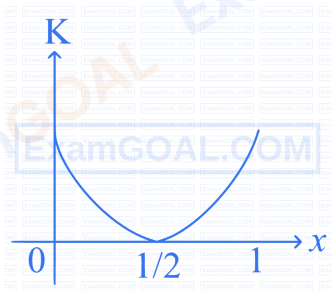
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Q19. JEE Main 2025 (Online) 24th January Evening Shift

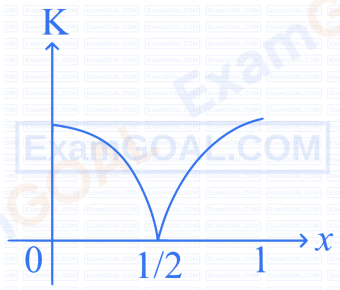
A particle oscillates along the x-axis according to the law, $x(t) = x_0 \sin^2\left(\frac{t}{2}\right)$ where $x_0 = 1\ \text{m}$. The kinetic energy (K) of the particle as a function of x is correctly represented by the graph

Open solution

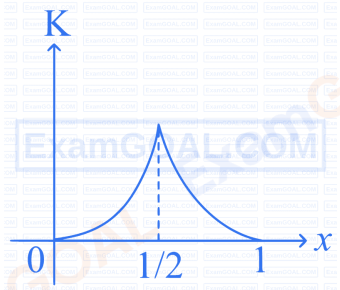
(A)



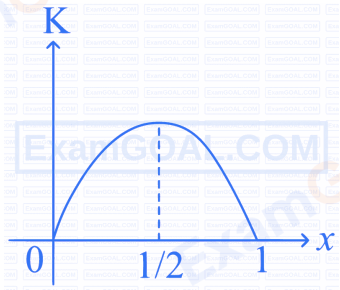
(B)



(C)



(D)



Q20. JEE Main 2025 (Online) 24th January Morning Shift

A particle is executing simple harmonic motion with time period 2 s and amplitude 1 cm . If D and d are the total distance and displacement covered by the particle in 12.5 s , then $\frac{D}{d}$ is

Open solution

(A) 10 (B) $\frac{16}{5}$ (C) 25 (D) $\frac{15}{4}$

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Q21. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

A light hollow cube of side length 10 cm and mass 10 g , is floating in water. It is pushed down and released to execute simple harmonic oscillations. The time period of oscillations is $y\pi \times 10^{-2}\,\mathrm{s}$, where the value of y is (Acceleration due to gravity, $g = 10\,\mathrm{m/s^2}$, density of water $= 10^3\,\mathrm{kg/m^3}$)

(A) 2 (B) 4 (C) 1 (D) 6

Q22. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : A simple pendulum is taken to a planet of mass and radius, 4 times and 2 times, respectively, that of the Earth. The time period of the pendulum remains same on earth and the planet.

Reason (R) : The mass of the pendulum remains unchanged at Earth and the other planet.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) (A) is false but (R) is true

(C) (A) is true but (R) is false

(D) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Q23. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

A simple pendulum doing small oscillations at a place R height above earth surface has time period of $T_1 = 4\,\mathrm{s}$. T_2 would be its time period if it is brought to a point which is at a height $2\mathrm{R}$ from earth surface. Choose the correct relation [R = radius of earth] :

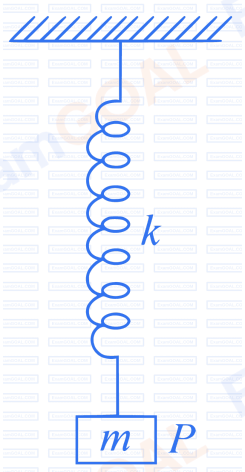
(A) $3\,\mathrm{T}_1 = 2\,\mathrm{T}_2$ (B) $\mathrm{T}_1 = \mathrm{T}_2$ (C) $2\,\mathrm{T}_1 = 3\,\mathrm{T}_2$ (D) $2\,\mathrm{T}_1 = \mathrm{T}_2$

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Q24. JEE Main 2024 (Online) 4th April Evening Shift

In simple harmonic motion, the total mechanical energy of given system is E. If mass of oscillating particle P is doubled then the new energy of the system for same amplitude is:

Open solution



(A) $E/\sqrt{2}$ (B) $2E$ (C) $E\sqrt{2}$ (D) E

Q25. JEE Main 2023 (Online) 15th April Morning Shift

In a linear Simple Harmonic Motion (SHM)

(A) Restoring force is directly proportional to the displacement.

(B) The acceleration and displacement are opposite in direction.

(C) The velocity is maximum at mean position.

(D) The acceleration is minimum at extreme points.

Choose the correct answer from the options given below:

(A) (A), (B) and (D) only (B) (C) and (D) only

(C) (A), (B) and (C) Only (D) (A), (C) and (D) only

Open solution

Q26. JEE Main 2023 (Online) 13th April Evening Shift

A particle executes SHM of amplitude A. The distance from the mean position when its kinetic energy becomes equal to its potential energy is :

(A) $\frac{1}{\sqrt{2}}A$ (B) $\frac{1}{2}A$ (C) $2\ \mathrm{A}$ (D) $\sqrt{2A}$

Open solution

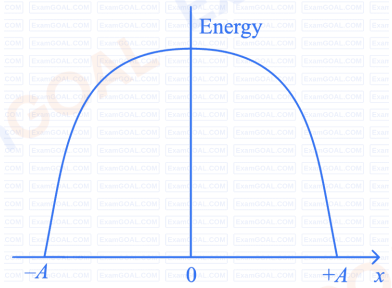
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Q27. JEE Main 2023 (Online) 13th April Morning Shift

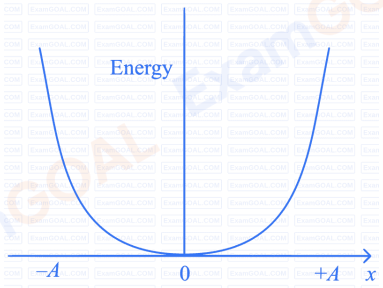
Which graph represents the difference between total energy and potential energy of a particle executing SHM vs its distance from mean position ?

Open solution

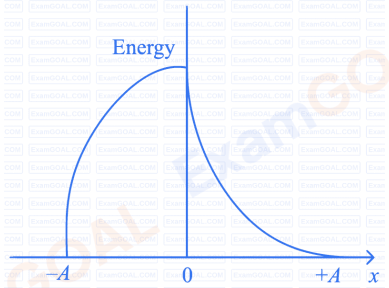
(A)



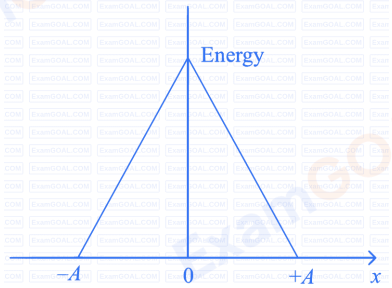
(B)



(C)



(D)



Q28. JEE Main 2023 (Online) 12th April Morning Shift

A particle is executing simple harmonic motion (SHM). The ratio of potential energy and kinetic energy of the particle when its displacement is half of its amplitude will be

(A) 1 : 1 (B) 1 : 4 (C) 2 : 1 (D) 1 : 3

Open solution

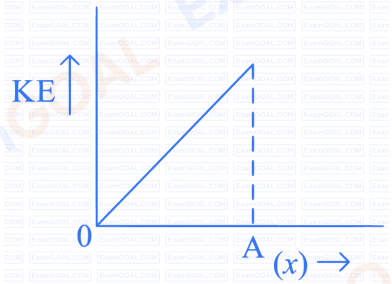
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Q29. JEE Main 2023 (Online) 11th April Morning Shift

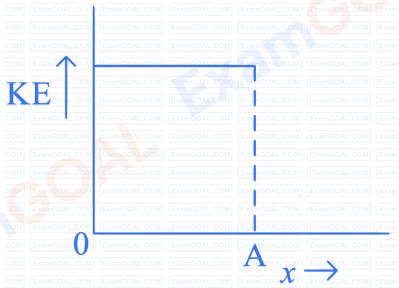
Open solution

The variation of kinetic energy (KE) of a particle executing simple harmonic motion with the displacement (x) starting from mean position to extreme position (A) is given by

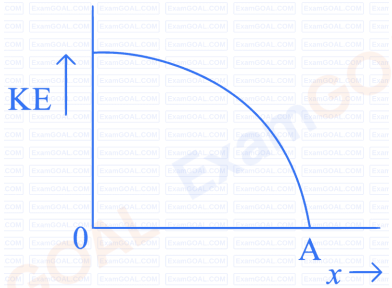
(A)



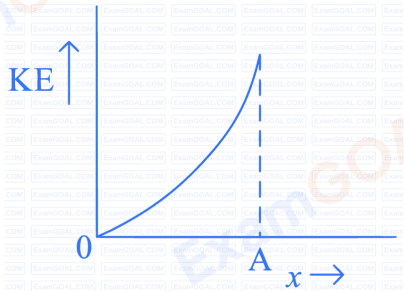
(B)



(C)



(D)



Q30. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

For a periodic motion represented by the equation

$y = \sin \omega t + \cos \omega t$

the amplitude of the motion is

(A) 1 (B) $\sqrt{2}$ (C) 0.5 (D) 2

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Q31. JEE Main 2023 (Online) 10th April Morning Shift

A particle executes S.H.M. of amplitude A along x-axis. At t = 0, the position of the particle is $x = \frac{A}{2}$ and it moves along positive x-axis. The displacement of particle in time t is $x = A\sin(wt + \delta)$, then the value of δ will be

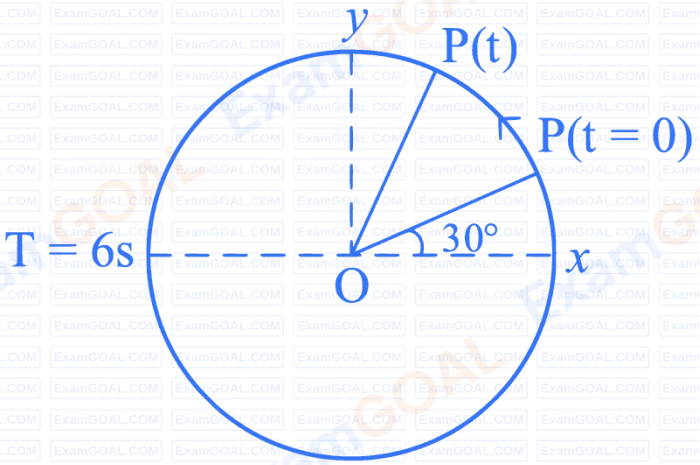
Open solution

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Q32. JEE Main 2023 (Online) 8th April Evening Shift

For particle P revolving round the centre O with radius of circular path r and angular velocity ω, as shown in below figure, the projection of OP on the x-axis at time t is

Open solution



(A) $x(t) = r\cos\left(\omega t - \frac{\pi}{6}\omega\right)$ (B) $x(t) = r\cos(\omega t)$

(C) $x(t) = r\cos\left(\omega t + \frac{\pi}{6}\right)$ (D) $x(t) = r\sin\left(\omega t + \frac{\pi}{6}\right)$

Q33. JEE Main 2023 (Online) 6th April Morning Shift

A mass m is attached to two springs as shown in figure. The spring constants of two springs are K_1 and K_2. For the frictionless surface, the time period of oscillation of mass m is :

Open solution



(A) $2\pi\sqrt{\frac{m}{K_1+K_2}}$ (B) $2\pi\sqrt{\frac{m}{K_1-K_2}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{K_1+K_2}{m}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{K_1-K_2}{m}}$

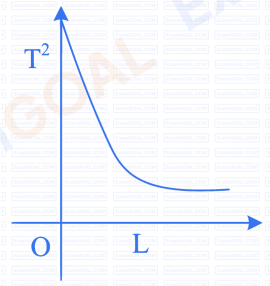
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Q34. JEE Main 2023 (Online) 1st February Evening Shift

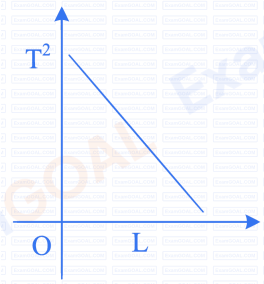
Choose the correct length (L) versus square of the time period (T^2) graph for a simple pendulum executing simple harmonic motion.

Open solution

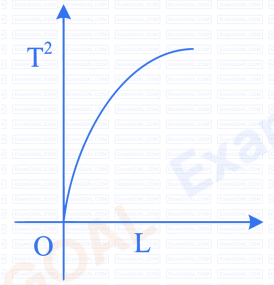
(A)



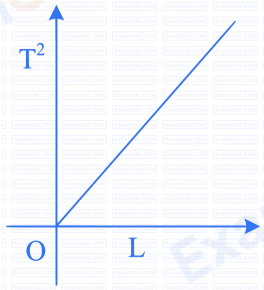
(B)



(C)



(D)



Q35. JEE Main 2023 (Online) 31st January Morning Shift

The maximum potential energy of a block executing simple harmonic motion is $25\ \mathrm{J}$. A is amplitude of oscillation. At $A/2$, the kinetic energy of the block is

(A) 9.75 J (B) 37.5 J (C) 18.75 J (D) 12.5 J

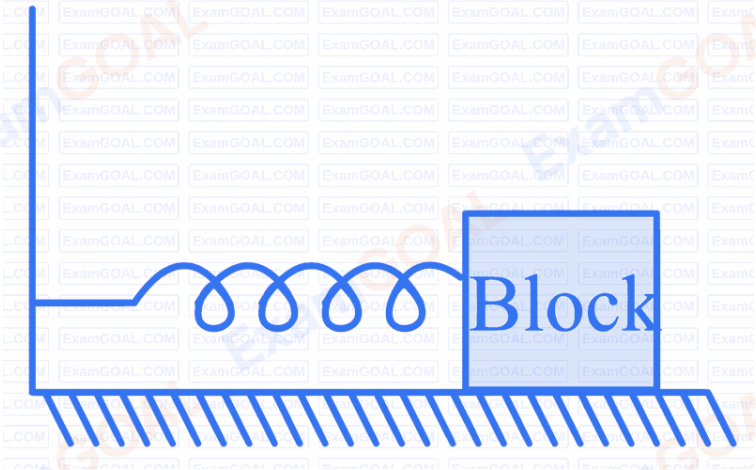
Open solution

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Q36. JEE Main 2023 (Online) 30th January Evening Shift

For a simple harmonic motion in a mass spring system shown, the surface is frictionless. When the mass of the block is $1\,\text{kg}$, the angular frequency is ω_1. When the mass block is $2\,\text{kg}$ the angular frequency is ω_2. The ratio ω_2/ω_1 is

Open solution



(A) $1/\sqrt{2}$ (B) $1/2$ (C) 2 (D) $\sqrt{2}$

Q37. JEE Main 2023 (Online) 25th January Evening Shift

A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If time taken by particle to go from $x = 0$ to $\frac{A}{2}$ is 2 s; then time taken by particle in going from $x = \frac{A}{2}$ to A is

(A) 4 s (B) 1.5 s (C) 3 s (D) 2 s

Open solution

Q38. JEE Main 2023 (Online) 25th January Morning Shift

T is the time period of simple pendulum on the earth's surface. Its time period becomes x T when taken to a height R (equal to earth's radius) above the earth's surface. Then, the value of x will be :

(A) 4 (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$

Open solution

Q39. JEE Main 2022 (Online) 30th July Evening Shift

An oscillating simple pendulum of time period T_o is placed in a lift which is accelerating upwards with $2.5\,\text{m/s}^2$. The time period of pendulum in the lift will be (if $g = 10\,\text{m/s}^2$)

(A) T_o (B) $\frac{2}{\sqrt{5}}T_o$ (C) $\frac{2}{\sqrt{3}}T_o$ (D) $\frac{T_o}{4}$

Open solution

Q40. JEE Main 2022 (Online) 29th July Morning Shift

The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by :

(A) $2\pi\sqrt{\text{L}/(\text{g}\cos\alpha)}$ (B) $2\pi\sqrt{\text{L}/(\text{g}\sin\alpha)}$

(C) $2\pi\sqrt{\text{L}/\text{g}}$ (D) $2\pi\sqrt{\text{L}/(\text{g}\tan\alpha)}$

Open solution

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Q41. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

Assume there are two identical simple pendulum clocks. Clock - 1 is placed on the earth and Clock - 2 is placed on a space station located at a height h above the earth surface. Clock - 1 and Clock - 2 operate at time periods 4 s and 6 s respectively. Then the value of h is -

(consider radius of earth $R_E = 6400\,\text{km}$ and g on earth $10\,\text{m/s}^2$)

(A) 1200 km (B) 1600 km (C) 3200 km (D) 4800 km

Q42. JEE Main 2022 (Online) 26th July Morning Shift

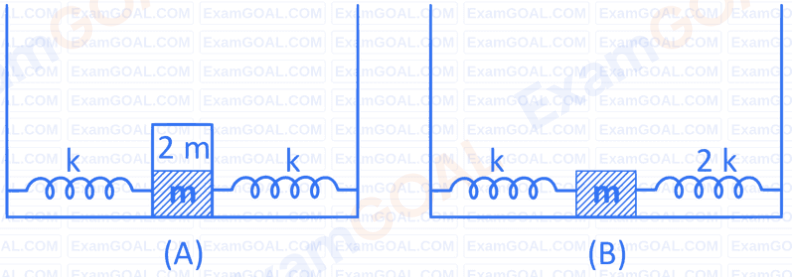
Open solution

When a particle executes Simple Harmonic Motion, the nature of graph of velocity as a function of displacement will be :

(A) Circular (B) Elliptical (C) Sinusoidal (D) Straight line

Q43. JEE Main 2022 (Online) 25th July Morning Shift

Open solution



In figure (A), mass '$2\,\mathrm{m}$' is fixed on mass 'm' which is attached to two springs of spring constant k. In figure (B), mass 'm' is attached to two springs of spring constant 'k' and '$2\mathrm{k}$'. If mass 'm' in (A) and in (B) are displaced by distance 'x' horizontally and then released, then time period T_1 and T_2 corresponding to (A) and (B) respectively follow the relation.

(A) $\frac{\mathrm{T}_1}{\mathrm{T}_2} = \frac{3}{\sqrt{2}}$ (B) $\frac{\mathrm{T}_1}{\mathrm{T}_2} = \sqrt{\frac{3}{2}}$

(C) $\frac{\mathrm{T}_1}{\mathrm{T}_2} = \sqrt{\frac{2}{3}}$ (D) $\frac{T_1}{T_2} = \frac{\sqrt{2}}{3}$

Q44. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

The motion of a simple pendulum executing S.H.M. is represented by the following equation.

$y = A\sin(\pi t + \phi)$, where time is measured in second. The length of pendulum is

(A) 97.23 cm (B) 25.3 cm (C) 99.4 cm (D) 406.1 cm

Q45. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Motion of a particle in x-y plane is described by a set of following equations $x = 4\sin\left(\frac{\pi}{2} - \omega t\right) m$ and $y = 4\sin(\omega t)\, m$. The path of the particle will be :

(A) circular (B) helical (C) parabolic (D) elliptical

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Q46. JEE Main 2022 (Online) 27th June Evening Shift

The equation of a particle executing simple harmonic motion is given by $x = \sin \pi \left(t + \frac{1}{3}\right) m$. At t = 1s, the speed of particle will be

(Given : π = 3.14)

Open solution

(A) 0 cm s^{-1} (B) 157 cm s^{-1} (C) 272 cm s^{-1} (D) 314 cm s^{-1}

Q47. JEE Main 2022 (Online) 27th June Morning Shift

The displacement of simple harmonic oscillator after 3 seconds starting from its mean position is equal to half of its amplitude. The time period of harmonic motion is :

Open solution

(A) 6 s (B) 8 s (C) 12 s (D) 36 s

Q48. JEE Main 2022 (Online) 26th June Morning Shift

Time period of a simple pendulum in a stationary lift is 'T'. If the lift accelerates with $\frac{g}{6}$ vertically upwards then the time period will be :

(Where g = acceleration due to gravity)

Open solution

(A) $\sqrt{\frac{6}{5}}T$ (B) $\sqrt{\frac{5}{6}}T$ (C) $\sqrt{\frac{6}{7}}T$ (D) $\sqrt{\frac{7}{6}}T$

Q49. JEE Main 2022 (Online) 24th June Evening Shift

Two massless springs with spring constants 2 k and 9 k, carry 50 g and 100 g masses at their free ends. These two masses oscillate vertically such that their maximum velocities are equal. Then, the ratio of their respective amplitudes will be :

Open solution

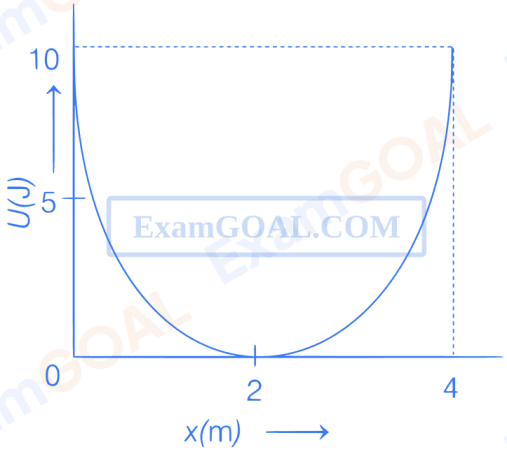
(A) 1 : 2 (B) 3 : 2 (C) 3 : 1 (D) 2 : 3

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Q50. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

A mass of 5 kg is connected to a spring. The potential energy curve of the simple harmonic motion executed by the system is shown in the figure. A simple pendulum of length 4 m has the same period of oscillation as the spring system. What is the value of acceleration due to gravity on the planet where these experiments are performed?



(A) 10 m/s^2 (B) 5 m/s^2 (C) 4 m/s^2 (D) 9.8 m/s^2

Q51. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

A bob of mass 'm' suspended by a thread of length l undergoes simple harmonic oscillations with time period T. If the bob is immersed in a liquid that has density $\frac{1}{4}$ times that of the bob and the length of the thread is increased by 1/3rd of the original length, then the time period of the simple harmonic oscillations will be :-

(A) T (B) $\frac{3}{2}$T (C) $\frac{3}{4}$T (D) $\frac{4}{3}$T

Q52. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

For a body executing S.H.M. :

(1) Potential energy is always equal to its K.E.

(2) Average potential and kinetic energy over any given time interval are always equal.

(3) Sum of the kinetic and potential energy at any point of time is constant.

(4) Average K.E. in one time period is equal to average potential energy in one time period.

Choose the most appropriate option from the options given below :

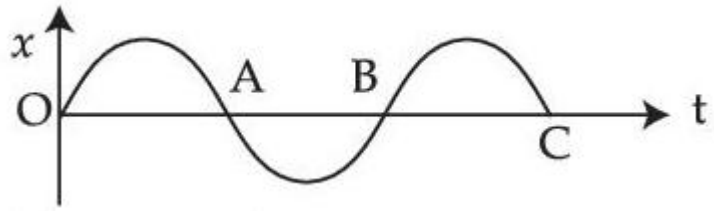
(A) (3) and (4) (B) only (3) (C) (2) and (3) (D) only (2)

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Q53. JEE Main 2021 (Online) 27th August Morning Shift

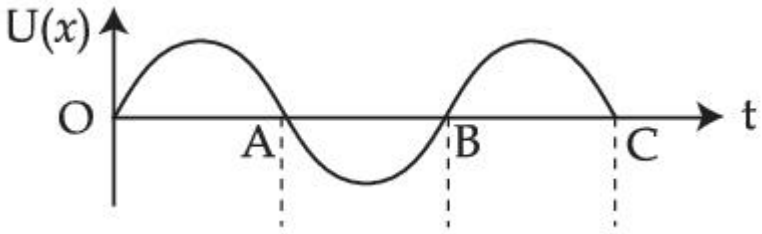
Open solution

The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure.

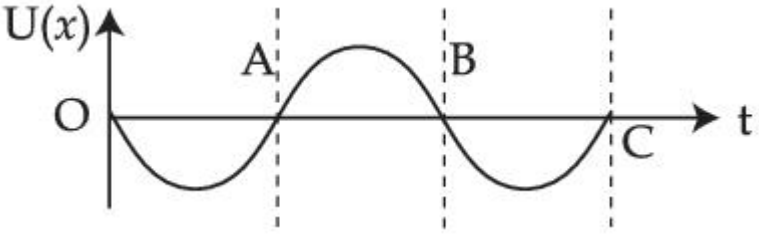


The potential energy U(x) versus time (t) plot of the particle is correctly shown in figure :

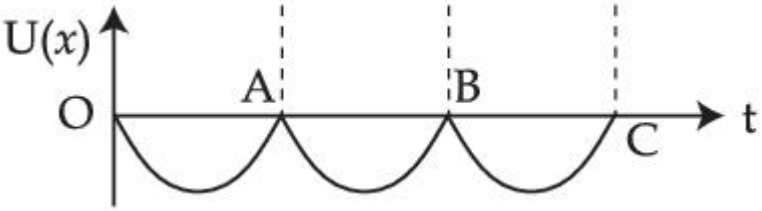
(A)



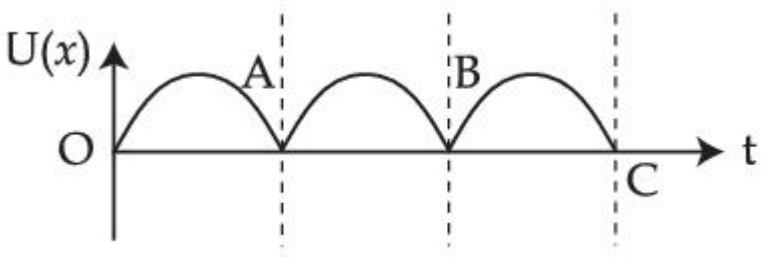
(B)



(C)



(D)



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Q54. JEE Main 2021 (Online) 27th July Evening Shift

An object of mass 0.5 kg is executing simple harmonic motion. It amplitude is 5 cm and time period (T) is 0.2 s. What will be the potential energy of the object at an instant $t = \frac{T}{4}s$ starting from mean position. Assume that the initial phase of the oscillation is zero.

Open solution

(A) 0.62 J (B) 6.2×10^{-3} J (C) 1.2×10^{3} J (D) 6.2×10^{3} J

Q55. JEE Main 2021 (Online) 27th July Morning Shift

A particle starts executing simple harmonic motion (SHM) of amplitude 'a' and total energy E. At any instant, its kinetic energy is $\frac{3E}{4}$ then its displacement 'y' is given by :

Open solution

(A) y = a (B) $y = \frac{a}{\sqrt{2}}$ (C) $y = \frac{a\sqrt{3}}{2}$ (D) $y = \frac{a}{2}$

Q56. JEE Main 2021 (Online) 25th July Evening Shift

In a simple harmonic oscillation, what fraction of total mechanical energy is in the form of kinetic energy, when the particle is midway between mean and extreme position.

Open solution

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Q57. JEE Main 2021 (Online) 22th July Evening Shift

T_0 is the time period of a simple pendulum at a place. if the length of the pendulum is reduced to $\frac{1}{16}$ times of its initial value, the modified time period is :

Open solution

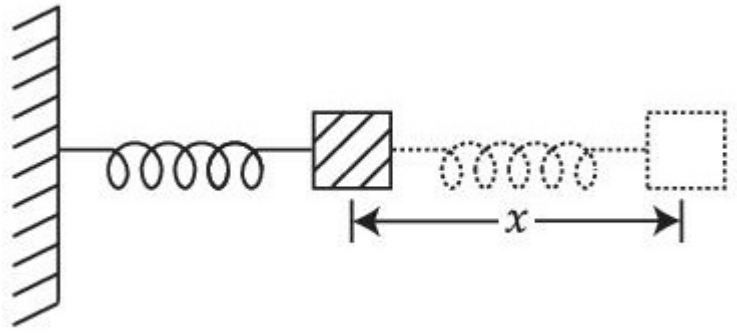
(A) $4\,T_0$ (B) $\frac{1}{4}\,T_0$ (C) T_0 (D) $8\pi\,T_0$

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Q58. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

The motion of a mass on a spring, with spring constant K is as shown in figure.



The equation of motion is given by
x(t) = A sinωt + B cosωt with $\omega = \sqrt{\frac{K}{m}}$

Suppose that at time t = 0, the position of mass is x(0) and velocity v(0), then its displacement can also be represented as x(t) = C cos($\omega t - \phi$), where C and ϕ are :

(A) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{2v(0)}\right)$

(B) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{x(0)\omega}{v(0)}\right)$

(C) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

(D) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1}\left(\frac{v(0)}{x(0)\omega}\right)$

Q59. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

A particle is making simple harmonic motion along the X-axis. If at a distances x_1 and x_2 from the mean position the velocities of the particle are v_1 and v_2 respectively. The time period of its oscillation is given as :

(A) $T = 2\pi\sqrt{\frac{x_2^2+x_1^2}{v_1^2-v_2^2}}$

(B) $T = 2\pi\sqrt{\frac{x_2^2+x_1^2}{v_1^2+v_2^2}}$

(C) $T = 2\pi\sqrt{\frac{x_2^2-x_1^2}{v_1^2+v_2^2}}$

(D) $T = 2\pi\sqrt{\frac{x_2^2-x_1^2}{v_1^2-v_2^2}}$

Q60. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is :

(A) cos(ωt) + cos(2ωt) + cos(3ωt)

(B) $\sin^2(\omega t)$

(C) sin(ωt) + cos(ωt)

(D) $3\cos\left(\frac{\pi}{4} - 2\omega t\right)$

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Q61. JEE Main 2021 (Online) 17th March Evening Shift

Two particles A and B of equal masses are suspended from two massless springs of spring constants K_1 and K_2 respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A and B is

(A) $\frac{K_1}{K_2}$ (B) $\sqrt{\frac{K_1}{K_2}}$ (C) $\frac{K_2}{K_1}$ (D) $\sqrt{\frac{K_2}{K_1}}$

Open solution

Q62. JEE Main 2021 (Online) 17th March Morning Shift

For what value of displacement the kinetic energy and potential energy of a simple harmonic oscillation become equal ?

(A) x = $\frac{A}{2}$ (B) x = $\pm$ A

(C) x = $\pm \frac{A}{\sqrt{2}}$ (D) x = 0

Open solution

Q63. JEE Main 2021 (Online) 16th March Morning Shift

Time period of a simple pendulum is T inside a lift when the lift is stationary. If the lift moves upwards with an acceleration g/2, the time period of pendulum will be :

(A) $\sqrt{3}T$ (B) $\sqrt{\frac{2}{3}}T$ (C) $\frac{T}{\sqrt{3}}$ (D) $\sqrt{\frac{3}{2}}T$

Open solution

Q64. JEE Main 2021 (Online) 26th February Evening Shift

A particle executes S.H.M., the graph of velocity as a function of displacement is :

(A) a parabola (B) a helix (C) an ellipse (D) a circle

Open solution

Q65. JEE Main 2021 (Online) 26th February Evening Shift

Given below are two statements :

Statement I : A second's pendulum has a time period of 1 second.

Statement II : It takes precisely one second to move between the two extreme positions.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Open solution

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Q66. JEE Main 2021 (Online) 26th February Morning Shift

Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance (R/2) from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period :

Open solution

(A) $2\pi\sqrt{\frac{R}{g}}$ (B) $\frac{g}{2\pi R}$

(C) $\frac{2\pi R}{g}$ (D) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$

Q67. JEE Main 2021 (Online) 26th February Morning Shift

If two similar springs each of spring constant K_1 are joined in series, the new spring constant and time period would be changed by a factor :

Open solution

(A) $\frac{1}{2}, 2\sqrt{2}$ (B) $\frac{1}{4}, 2\sqrt{2}$ (C) $\frac{1}{2}, \sqrt{2}$ (D) $\frac{1}{4}, \sqrt{2}$

Q68. JEE Main 2021 (Online) 25th February Evening Shift

Y = A sin(ωt + ϕ_0) is the time-displacement equation of a SHM. At t = 0 the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x-direction. Then the initial phase angle ϕ_0 will be:

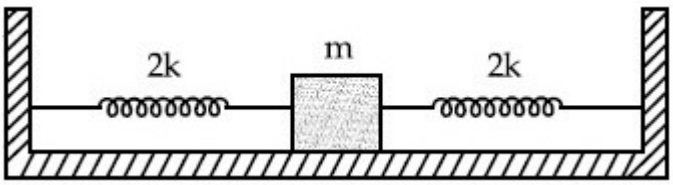
Open solution

(A) $\frac{5\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{6}$

Q69. JEE Main 2021 (Online) 25th February Evening Shift

Two identical springs of spring constant '2k' are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. The time period of oscillations of this system is :

Open solution



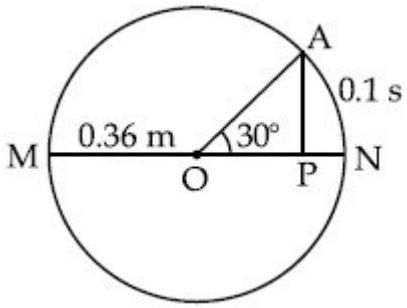
(A) $2\pi\sqrt{\frac{m}{k}}$ (B) $\pi\sqrt{\frac{m}{k}}$ (C) $2\pi\sqrt{\frac{m}{2k}}$ (D) $\pi\sqrt{\frac{m}{2k}}$

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Q70. JEE Main 2021 (Online) 25th February Evening Shift

The point A moves with a uniform speed along the circumference of a circle of radius 0.36 m and covers 30° in 0.1 s. The perpendicular projection 'P' from 'A' on the diameter MN represents the simple harmonic motion of 'P'. The restoration force per unit mass when P touches M will be :

Open solution



(A) 9.87 N (B) 0.49 N (C) 50 N (D) 100 N

Q71. JEE Main 2021 (Online) 25th February Morning Shift

If the time period of a two meter long simple pendulum is 2s, the acceleration due to gravity at the place where pendulum is executing S.H.M. is :

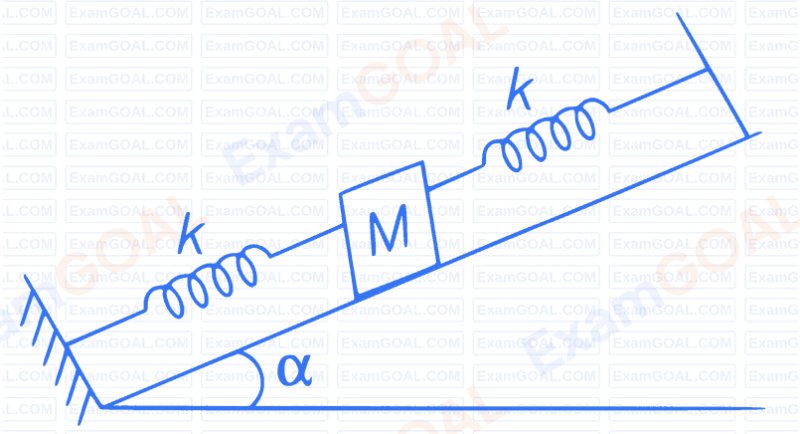
Open solution

(A) 16 m/s^2 (B) $2\pi^2$ ms^{-2} (C) π^2 ms^{-2} (D) 9.8 ms^{-2}

Q72. JEE Main 2021 (Online) 24th February Evening Shift

In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k, the frequency of oscillation of given body is :

Open solution



(A) $\frac{1}{2\pi}\sqrt{\frac{2k}{Mg\sin\alpha}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{k}{Mg\sin\alpha}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{k}{2M}}$

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Q73. JEE Main 2021 (Online) 24th February Evening Shift

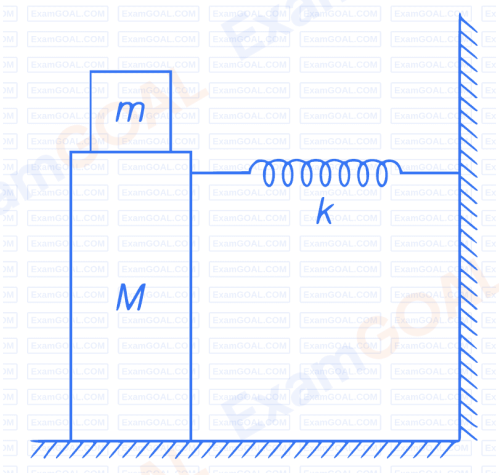
When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement is :

(A) circular (B) straight line (C) parabolic (D) elliptical

Open solution

Q74. JEE Main 2021 (Online) 24th February Morning Shift

In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k. The mass oscillates on a frictionless surface with time period T and amplitude A. When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillations will be :



(A) $A\sqrt{\frac{M}{M-m}}$ (B) $A\sqrt{\frac{M-m}{M}}$ (C) $A\sqrt{\frac{M+m}{M}}$ (D) $A\sqrt{\frac{M}{M+m}}$

Open solution

Q75. JEE Main 2020 (Online) 6th September Evening Slot

When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by

$y(t) = y_0 \sin^2 \omega t$, where 'y' is measured from the lower end of unstretched spring. Then ω is:

(A) $\sqrt{\frac{g}{y_0}}$ (B) $\frac{1}{2}\sqrt{\frac{g}{y_0}}$

(C) $\sqrt{\frac{2g}{y_0}}$ (D) $\sqrt{\frac{g}{2y_0}}$

Open solution

Q76. JEE Main 2020 (Online) 5th September Evening Slot

A ring is hung on a nail. It can oscillate, without slipping or sliding

(i) in its plane with a time period T_1 and,

(ii) back and forth in a direction perpendicular to its plane,

with a period T_2. The ratio $\frac{T_1}{T_2}$ will be :

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{2}{3}$ (D) $\frac{3}{\sqrt{2}}$

Open solution

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Q77. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

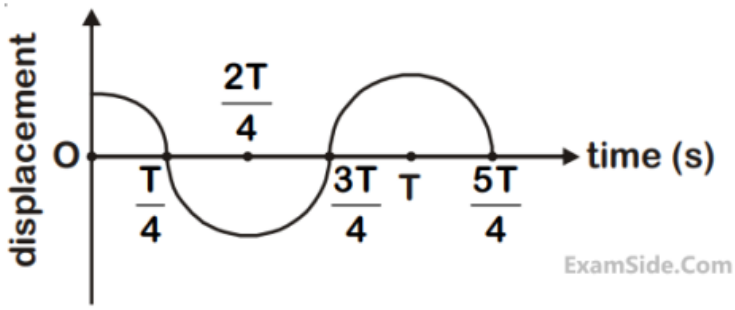
A block of mass m attached to a massless spring is performing oscillatory motion of amplitude 'A' on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system become fA. The value of f is :

(A) 1 (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Q78. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

The displacement time graph of a particle executing S.H.M is given in figure :
(sketch is schematic and not to scale)



Which of the following statements is/are true for this motion?
(A) The force is zero at t = $\frac{3T}{4}$
(B) The acceleration is maximum at t = T
(C) The speed is maximum at t = $\frac{T}{4}$
(D) The P.E. is equal to K.E. of the oscillation at t = $\frac{T}{2}$

(A) (B), (C) and (D) **(B)** (A), (B) and (C) **(C)** (A) and (D) **(D)** (A), (B) and (D)

Q79. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is 1/16 th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is :

(A) $2T\sqrt{\frac{1}{10}}$ (B) $4T\sqrt{\frac{1}{14}}$ (C) $4T\sqrt{\frac{1}{15}}$ (D) $2T\sqrt{\frac{1}{14}}$

Q80. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

A simple harmonic motion is represented by :

$y = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)$ cm

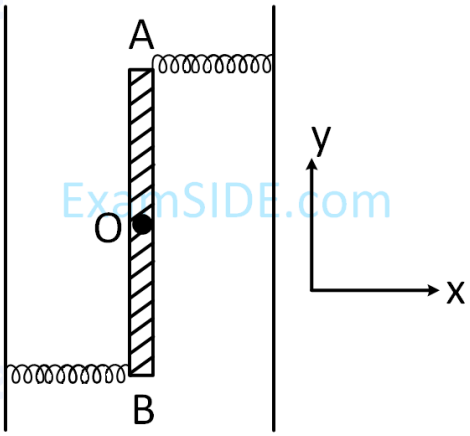
The amplitude and time period of the motion are :

(A) 10 cm, $\frac{3}{2}$ s (B) 5 cm, $\frac{2}{3}$ s (C) 5 cm, $\frac{3}{2}$ s (D) 10 cm, $\frac{2}{3}$ s

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Q81. JEE Main 2019 (Online) 12th January Morning Slot

Two light identical springs of spring constant k are attached horizontally at the two ends of a uniform horizontal rod AB of length ℓ and mass m. The rod is pivoted at its centre 'O' and can rotate freely in horizontal plane. The other ends of the two springs are fixed to rigid supports as shown in figure. The rod is gently pushed through a small angle and released. The frequency of resulting oscillation is :



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Open solution

(A) $\frac{1}{2\pi}\sqrt{\frac{3k}{m}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{6k}{m}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

Q82. JEE Main 2019 (Online) 11th January Evening Slot

A pendulum is executing simple harmonic motion and its maximum kinetic energy is K_1. If the length of the pendulum is doubled and it performs simple harmonic motion with the same amplitude as in the first case, its maximum kinetic energy is K_2. Then :

Open solution

(A) $K_2 = \frac{K_1}{2}$ (B) $K_2 = 2K_1$ (C) $K_2 = K_1$ (D) $K_2 = \frac{K_1}{4}$

Q83. JEE Main 2019 (Online) 11th January Evening Slot

The mass and the diameter of a planet are three times the respective values for the Earth. The period of oscillation of simple pendulum on the Earth is 2 s. The period of oscillation of the same pendulum on the planet would be :

Open solution

(A) $\frac{\sqrt{3}}{2}$ s (B) $\frac{3}{2}$ s (C) $\frac{2}{\sqrt{3}}$ s (D) $2\sqrt{3}$ s

Q84. JEE Main 2019 (Online) 11th January Morning Slot

A particle undergoing simple harmonic motion has time dependent displacement given by $x(t) = A\sin\frac{\pi t}{90}$. The ratio of kinetic to potential energy of this particle at t = 210 s will be:

Open solution

(A) $\frac{1}{9}$ (B) 3 (C) 2 (D) 1

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Q85. JEE Main 2019 (Online) 10th January Evening Slot

A cylindrical plastic bottle of negligible mass is filled with 310 ml of water and left floating in a pond with still water. If pressed downward slightly and released, it starts performing simple harmonic motion at angular frequency ω. If the radius of the bottle is 2.5 cm then ω is close to – (density of water = 10^3 kg/m^3).

(A) 2.50 rad s^{-1} (B) 3.75 rad s^{-1} (C) 5.00 rad s^{-1} (D) 7.90 rad s^{-1}

Open solution

Q86. JEE Main 2019 (Online) 10th January Evening Slot

A particle executes simple harmonic motion with an amplitude of 5 cm. When the particle is at 4 cm from the mean position, the magnitude of its velocity in SI units is equal to that of its acceleration. Then, its periodic time in seconds is -

(A) $\frac{4\pi}{3}$ (B) $\frac{3}{8}\pi$ (C) $\frac{7}{3}\pi$ (D) $\frac{8\pi}{3}$

Open solution

Q87. JEE Main 2019 (Online) 9th January Evening Slot

A particle is executing simple harmonic motion (SHM) of amplitude A, along the x-axis, about x = 0. When its potential Energy (PE) equals kinetic energy (KE), the position of the particle will be :

(A) $\frac{A}{2}$ (B) $\frac{A}{2\sqrt{2}}$ (C) $\frac{A}{\sqrt{2}}$ (D) A

Open solution

Q88. JEE Main 2019 (Online) 9th January Evening Slot

A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to :

(A) 0.77 (B) 0.57 (C) 0.37 (D) 0.17

Open solution

Q89. JEE Main 2018 (Online) 16th April Morning Slot

An oscillator of mass M is at rest in its equilibrium position in a potential $V = \frac{1}{2}k(x - X)^2$. A particle of mass m comes from right with speed u and collides completely inelastically with M and sticks to it. This process repeats every time the oscillator crosses its equilibrium position. The amplitude of oscillations after 13 collisions is : (M = 10, m = 5, u = 1, k = 1)

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\sqrt{\frac{3}{5}}$

Open solution

Q90. JEE Main 2018 (Online) 16th April Morning Slot

A particle executes simple harmonic motion and is located at x = a, b and c at times t_0, $2t_0$ and $3t_0$ respectively. The frequency of the oscillation is :

(A) $\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+c}{2b}\right)$ (B) $\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+b}{2c}\right)$

(C) $\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{2a+3c}{b}\right)$ (D) $\frac{1}{2\pi t_0}\cos^{-1}\left(\frac{a+2b}{3c}\right)$

Open solution

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Q91. JEE Main 2018 (Offline)

A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of 10^{12}/sec. What is the force constant of the bonds connecting one atom with the other? (Mole wt. of silver = 108 and Avogadro number = 6.02×10^{23} gm mole^{-1})

Open solution

(A) 5.5 N/m (B) 6.4 N/m (C) 7.1 N/m (D) 2.2 N/m

Q92. JEE Main 2018 (Online) 15th April Evening Slot

Two simple harmonic motions, as shown below, are at right angles. They are combined to form Lissajous figures.
x(t) = A sin (at + δ)
y(t) = B sin (bt)

Open solution

Identify the **correct** match below.

(A) **Parameters** A $\neq$ B, a = b; δ = 0;
Curve Parabola

(B) **Parameters** A = B, a = b; $\delta = \frac{\pi}{2}$
Curve Line

(C) **Parameters** A $\neq$ B, a = b; $\delta = \frac{\pi}{2}$
Curve Ellipse

(D) **Parameters** A = B, a = 2b; $\delta = \frac{\pi}{2}$
Curve Circle

Q93. JEE Main 2017 (Online) 9th April Morning Slot

In an experiment to determine the period of a simple pendulum of length 1 m, it is attached to different spherical bobs of radii r_1 and r_2 . The two spherical bobs have uniform mass distribution. If the relative difference in the periods, is found to be 5×10^{-4} s, the difference in radii, $|r_1 - r_2|$ is best given by :

Open solution

(A) 1 cm (B) 0.1 cm (C) 0.5 cm (D) 0.01 cm

Q94. JEE Main 2017 (Online) 8th April Morning Slot

A 1 kg block attached to a spring vibrates with a frequency of 1 Hz on a frictionless horizontal table. Two springs identical to the original spring are attached in parallel to an 8 kg block placed on the same table. So, the frequency of vibration of the 8 kg block is :

Open solution

(A) $\frac{1}{4}Hz$ (B) $\frac{1}{2\sqrt{2}}Hz$ (C) $\frac{1}{2}Hz$ (D) $2\ Hz$

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Q95. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

The ratio of maximum acceleration to maximum velocity in a simple harmonic motion is 10 s^{-1} . At, t = 0 the displacement is 5 m. What is the maximum acceleration ? The initial phase is $\frac{\pi}{4}$.

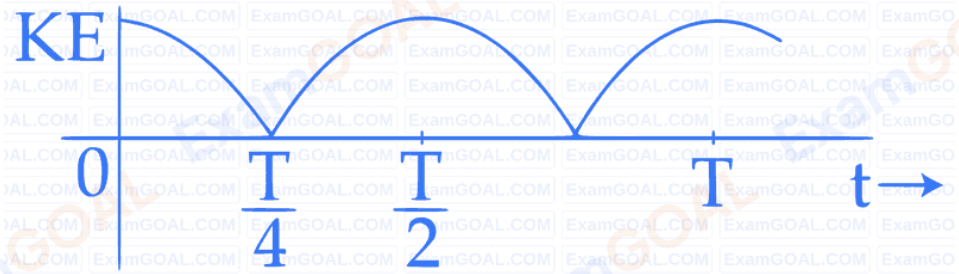
(A) 500 m/s^2 (B) 500 $\sqrt{2}m/\ s^2$ (C) 750 m/s^2 (D) 750 $\sqrt{2}$m / s^2

Q96. JEE Main 2017 (Offline)

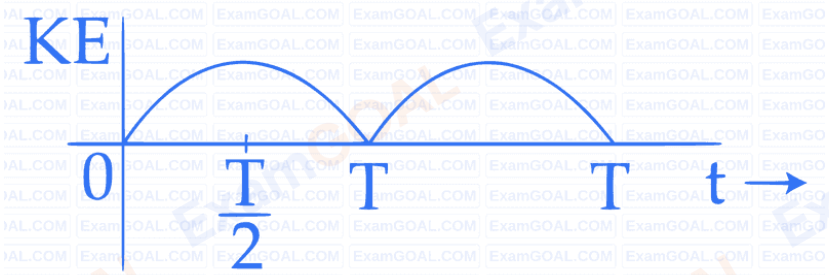
Open solution

A particle is executing simple harmonic motion with a time period T. At time t = 0, it is at its position of equilibrium. The kinetic energy – time graph of the particle will look like:

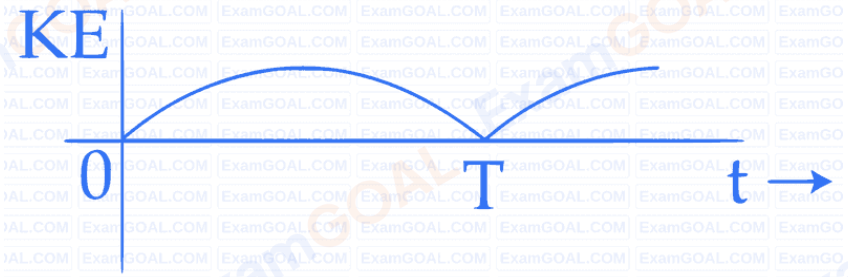
(A)



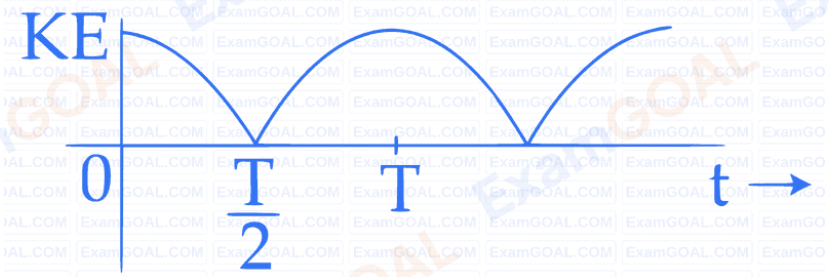
(B)



(C)



(D)



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Q97. JEE Main 2016 (Online) 10th April Morning Slot

In an engine the piston undergoes vertical simple harmonic motion with amplitude 7 cm. A washer rests on top of the piston and moves with it. The motor speed is slowly increased. The frequency of the piston at which the washer no longer stays in contact with the piston, is close to :

Open solution

(A) 0.1 Hz (B) 1.2 Hz (C) 0.7 Hz (D) 1.9 Hz

Q98. JEE Main 2016 (Online) 9th April Morning Slot

Two particles are performing simple harmonic motion in a straight line about the same equilibrium point. The amplitude and time period for both particles are same and equal to A and T, respectively. At time t = 0 one particle has displacement A while the other one has displacement $\frac{-A}{2}$ and they are moving towards each other. If they cross each other at time t, then t is :

Open solution

(A) $\frac{T}{6}$ (B) $\frac{5T}{6}$ (C) $\frac{T}{3}$ (D) $\frac{T}{4}$

Q99. JEE Main 2016 (Offline)

A particle performs simple harmonic motion with amplitude A. Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is:

Open solution

(A) $A\sqrt{3}$ (B) $\frac{7A}{3}$ (C) $\frac{A}{3}\sqrt{41}$ (D) $3A$

Q100. JEE Main 2015 (Online) 11th April Morning Slot

A cylindrical block of wood (density = 650 kg m^{-3}), of base area 30 cm^2 and height 54 cm, floats in a liquid of density 900 kg m^{-3}. The block is depressed slightly and then released. The time period of the resulting oscillations of the block would be equal to that of a simple pendulum of length (nearly) :

Open solution

(A) 65 cm (B) 52 cm (C) 39 cm (D) 26 cm

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Q101. JEE Main 2015 (Online) 10th April Morning Slot

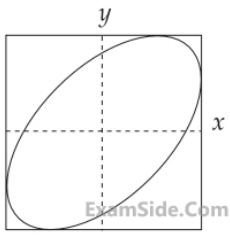
Open solution

x and y displacements of a particle are given as

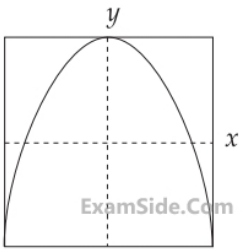
$x(t) = a \sin\omega t$ and $y(t) = a \sin 2\omega t$.

Its trajectory will look like :

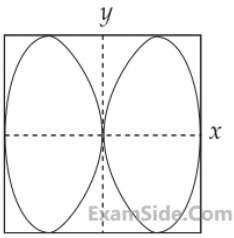
(A)



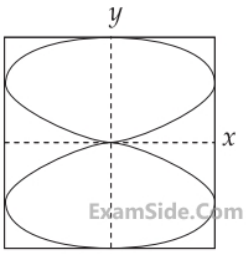
(B)



(C)



(D)



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Q102. JEE Main 2015 (Offline)

Open solution

A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to T_M. If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to :
($g =$ gravitational acceleration)

(A) $\left[1-\left(\frac{T_M}{T}\right)^2\right]\frac{A}{Mg}$

(B) $\left[1-\left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$

(C) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{A}{Mg}$

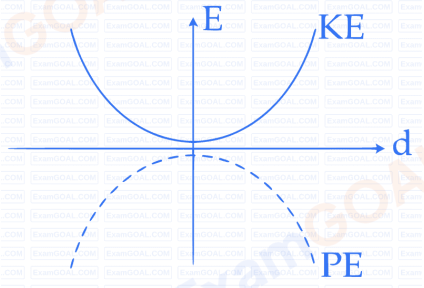
(D) $\left[\left(\frac{T_M}{T}\right)^2-1\right]\frac{Mg}{A}$

Q103. JEE Main 2015 (Offline)

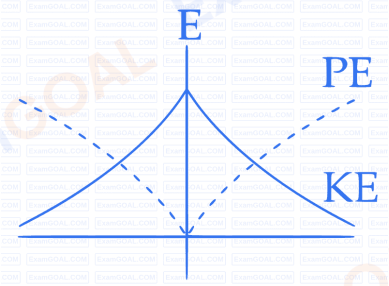
Open solution

For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement d. Which one of the following represents these correctly?
$(graphs\ are\ schematic\ and\ not\ drawn\ to\ scale)$

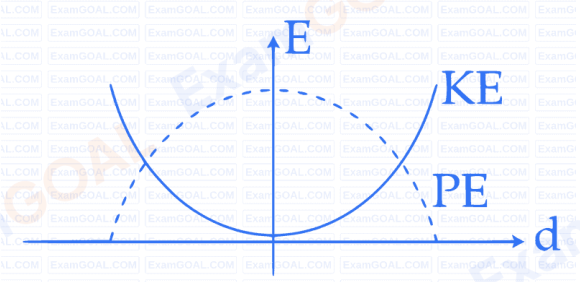
(A)



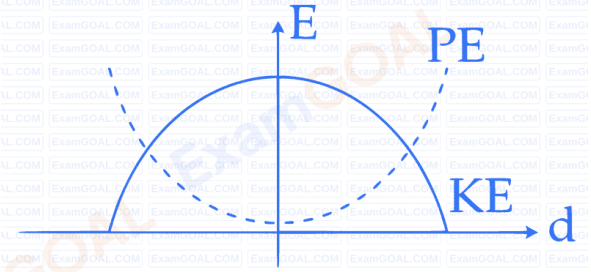
(B)



(C)



(D)



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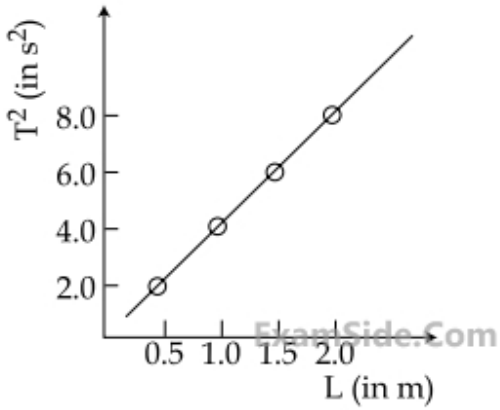
Q104. JEE Main 2014 (Online) 19th April Morning Slot

A body is in simple harmonic motion with time period half second (T = 0.5 s) and amplitude one cm (A = 1 cm). Find the average velocity in the interval in which it moves from equilibrium position to half of its amplitude.

Open solution

(A) 4 cm/s (B) 6 cm/s (C) 12 cm/s (D) 16 cm/s

Q105. JEE Main 2014 (Online) 19th April Morning Slot



Open solution

In an experiment for determining the gravitational acceleration g of a place with the help of a simple pendulum, the measured time period square is plotted against the string length of the pendulum in the figure.

what is the value of g at the place ?

(A) 9.81 m/s^2 (B) 9.87 m/s^2 (C) 9.91 m/s^2 (D) 10.0 m/s^2

Q106. JEE Main 2014 (Online) 12th April Morning Slot

Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants ?

Open solution

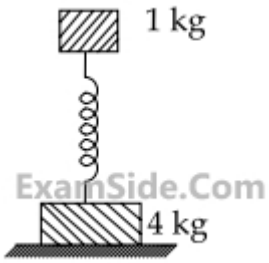
(A) $a + bx - cx^2$ (B) bx^2 (C) $a - bx + cx^2$ (D) $-bx$

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Q107. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Two bodies of masses 1 kg and 4 kg are connected to a vertical spring, as shown in the figure. The smaller mass executes simple harmonic motion of angular frequency 25 rad /s, and amplitude 1.6 cm while the bigger mass remains stationary on the ground. The maximum force exerted by the system on the floor is (take g = 10 ms $^{-2}$).



(A) 20 N (B) 10 N (C) 60 N (D) 40 N

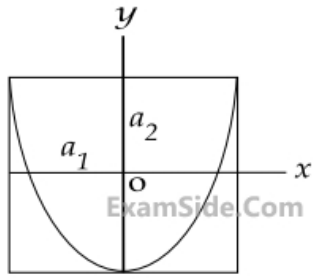
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Q108. JEE Main 2014 (Online) 9th April Morning Slot

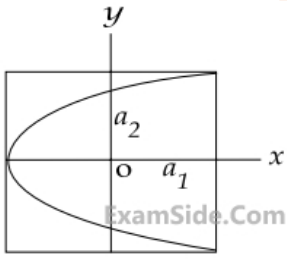
Open solution

A particle which is simultaneously subjected to two perpendicular simple harmonic motions represented by; $x = a_1 \cos \omega t$ and $y = a_2 \cos 2\omega t$ traces a curve given by :

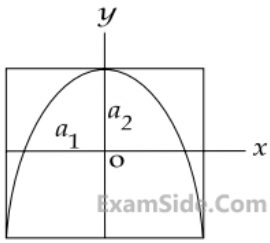
(A)



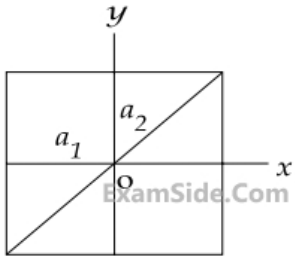
(B)



(C)



(D)



Q109. JEE Main 2014 (Offline)

Open solution

A particle moves with simple harmonic motion in a straight line. In first τs, after starting from rest it travels a distance a, and in next τs it travels $2a$, in same direction, then:

(A) amplitude of motion is $3a$

(B) time period of oscillations is 8τ

(C) amplitude of motion is $4a$

(D) time period of oscillations is 6τ

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Q110. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. When the cylinder is given a downward push and released, it starts oscillating vertically with a small amplitude. The time period T of the oscillations of the cylinder will be :

(A) Smaller than $2\pi\left[\frac{M}{(k+A\sigma g)}\right]^{1/2}$

(B) $2\pi\sqrt{\frac{M}{k}}$

(C) Larger than $2\pi\left[\frac{M}{(k+A\sigma g)}\right]^{1/2}$

(D) $2\pi\left[\frac{M}{(k+A\sigma g)}\right]^{1/2}$

Q111. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

Bob of a simple pendulum of length l is made of iron. The pendulum is oscillating over a horizontal coil carrying direct current. If the time period of the pendulum is T then :

(A) $T < 2\pi\sqrt{\frac{l}{g}}$ and damping is smaller than in air alone.

(B) $T = 2\pi\sqrt{\frac{l}{g}}$ and damping is larger than in air alone.

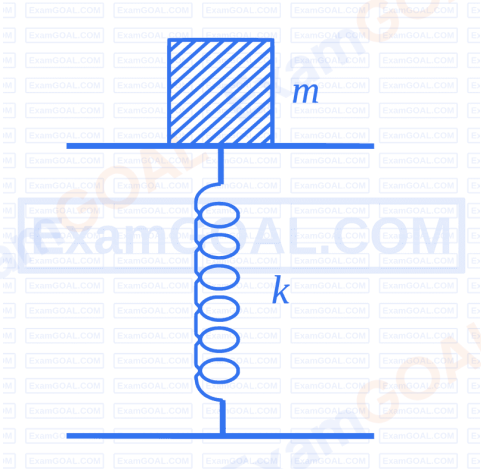
(C) $T > 2\pi\sqrt{\frac{l}{g}}$ and damping is smaller than in air alone.

(D) $T < 2\pi\sqrt{\frac{l}{g}}$ and damping is larger than in air alone.

Q112. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

A mass $m = 1.0$ kg is put on a flat pan attached to a vertical spring fixed on the ground. The mass of the spring and the pan is negligible. When pressed slightly and released, the mass executes simple harmonic motion. The spring constant is 500 N/m. What is the amplitude A of the motion, so that the mass m tends to get detached from the pan? (Take $g = 10$ m/s^2). The spring is stiff enough so that it does not get distorted during the motion.



(A) A $\geq$ 2.0 cm

(B) A = 2.0 cm

(C) A < 2.0 cm

(D) A = 1.5 cm

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Q113. JEE Main 2013 (Online) 9th April Morning Slot

Two simple pendulums of length 1 m and 4 m respectively are both given small displacement in the same direction at the same instant. They will be again in phase after the shorter pendulum has completed number of oscillations equal to:

Open solution

(A) 2 (B) 7 (C) 5 (D) 3

Q114. JEE Main 2013 (Offline)

An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M. The piston and the cylinder have equal cross sectional area A. When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0. The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency

Open solution

(A) $\frac{1}{2\pi}\frac{A\gamma P_0}{V_0 M}$ (B) $\frac{1}{2\pi}\frac{V_0 M P_0}{A^2\gamma}$

(C) $\frac{1}{2\pi}\sqrt{\frac{A^2\gamma P_0}{V_0 M}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{MV_0}{A\gamma P_0}}$

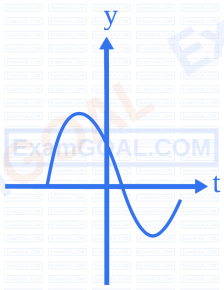
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Q115. AIEEE 2012 (Online) 19th May

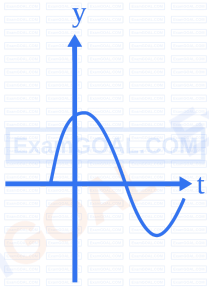
Open solution

The displacement $y(t) = A\sin(\omega t + \phi)$ of a pendulum for $\phi = \frac{2\pi}{3}$ is correctly represented by

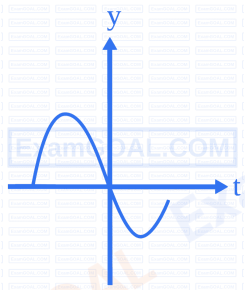
(A)



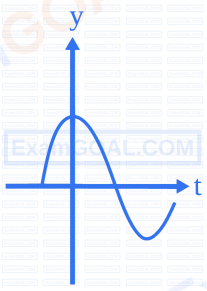
(B)



(C)



(D)

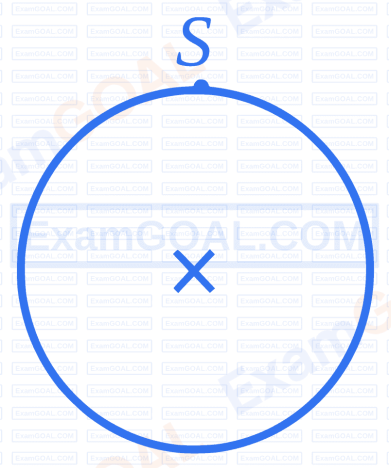


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Q116. AIEEE 2012 (Online) 19th May

A ring is suspended from a point S on its rim as shown in the figure. When displaced from equilibrium, it oscillates with time period of 1 second. The radius of the ring is (take $g = \pi^2$)

Open solution



(A) $0.15\ \text{m}$ (B) $1.5\ \text{m}$ (C) $1.0\ \text{m}$ (D) $0.5\ \text{m}$

Q117. AIEEE 2011

Two particles are executing simple harmonic motion of the same amplitude A and frequency ω along the x-axis. Their mean position is separated by distance $X_0\ (X_0 > A)$. If the maximum separation between them is $(X_0 + A)$, the phase difference between their motion is:

Open solution

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

Q118. AIEEE 2011

A mass M, attached to a horizontal spring, executes $S.H.M.$ with amplitude A_1. When the mass M passes through its mean position then a smaller mass m is placed over it and both of them move together with amplitude A_2. The ratio of $\left(\frac{A_1}{A_2}\right)$ is :

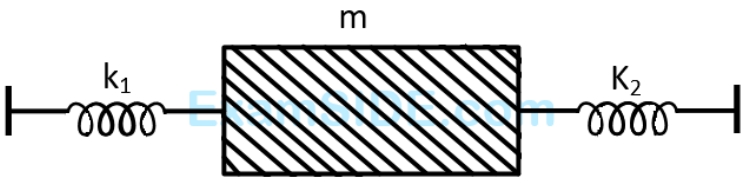
Open solution

(A) $\frac{M+m}{M}$ (B) $\left(\frac{M}{M+m}\right)^{\frac{1}{2}}$
(C) $\left(\frac{M+m}{M}\right)^{\frac{1}{2}}$ (D) $\frac{M}{M+m}$

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Q119. AIEEE 2007

Two springs, of force constant k_1 and k_2 are connected to a mass m as shown. The frequency of oscillation of the mass is f. If both k_1 and k_2 are made four times their original values, the frequency of oscillation becomes



Open solution

(A) $2f$ (B) $f/2$ (C) $f/4$ (D) $4f$

Q120. AIEEE 2007

A particle of mass m executes simple harmonic motion with amplitude a and frequency v. The average kinetic energy during its motion from the position of equilibrium to the end is

Open solution

(A) $2\pi^2\, ma^2v^2$ (B) $\pi^2\, ma^2v^2$ (C) $\frac{1}{4}\, ma^2v^2$ (D) $4\pi^2 ma^2v^2$

Q121. AIEEE 2007

The displacement of an object attached to a spring and executing simple harmonic motion is given by $x = 2 \times 10^{-2}\ cos\ \pi t$ metre. The time at which the maximum speed first occurs is

Open solution

(A) $0.25\ s$ (B) $0.5\ s$ (C) $0.75\ s$ (D) $0.125\ s$

Q122. AIEEE 2007

A point mass oscillates along the x-axis according to the law $x = x_0\ \cos(\omega t - \pi/4)$. If the acceleration of the particle is written as $a = A\ \cos(\omega t + \delta)$, then

Open solution

(A) $A = x_0\omega^2,\ \delta = 3\pi/4$ (B) $A = x_0,\ \delta = -\pi/4$
(C) $A = x_0\omega^2,\ \delta = \pi/4$ (D) $A = x_0\omega^2,\ \delta = -\pi/4$

Q123. AIEEE 2006

The maximum velocity of a particle, executing simple harmonic motion with an amplitude $7\ mm$, is $4.4\ m/s$. The period of oscillation is

Open solution

(A) $0.01\ s$ (B) $10\ s$ (C) $0.1\ s$ (D) $100\ s$

Q124. AIEEE 2006

Starting from the origin a body oscillates simple harmonically with a period of $2\ s$. After what time will its kinetic energy be 75% of the total energy?

Open solution

(A) $\frac{1}{6}s$ (B) $\frac{1}{4}s$ (C) $\frac{1}{3}s$ (D) $\frac{1}{12}s$

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Q125. AIEEE 2005

Two simple harmonic motions are represented by the equations $y_1 = 0.1 \sin\left(100\pi t + \frac{\pi}{3}\right)$ and $y_2 = 0.1 \cos \pi t$. The phase difference of the velocity of particle 1 with respect to the velocity of particle 2 is

(A) $\frac{\pi}{3}$ (B) $\frac{-\pi}{6}$ (C) $\frac{\pi}{6}$ (D) $\frac{-\pi}{3}$

Open solution

Q126. AIEEE 2005

The function $\sin^2(\omega t)$ represents

(A) a periodic, but not SHM with a period $\frac{\pi}{\omega}$

(B) a periodic, but not SHM with a period $\frac{2\pi}{\omega}$

(C) a SHM with a period $\frac{\pi}{\omega}$

(D) a SHM with a period $\frac{2\pi}{\omega}$

Open solution

Q127. AIEEE 2005

The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out, the time period of oscillation would

(A) first decrease and then increase to the original value

(B) first increase and then decrease to the original value

(C) increase towards a saturation value

(D) remain unchanged

Open solution

Q128. AIEEE 2005

If a simple harmonic motion is represented by $\frac{d^2x}{dt^2} + \alpha x = 0$. its time period is

(A) $\frac{2\pi}{\sqrt{\alpha}}$ (B) $\frac{2\pi}{\alpha}$ (C) $2\pi\sqrt{\alpha}$ (D) $2\pi\alpha$

Open solution

Q129. AIEEE 2004

The bob of a simple pendulum executes simple harmonic motion in water with a period t, while the period of oscillation of the bob is t_0 in air. Neglecting frictional force of water and given that the density of the bob is $(4/3) \times 1000\ kg/m^3$. What relationship between t and t_0 is true

(A) $t = 2t_0$ (B) $t = t_0/2$ (C) $t = t_0$ (D) $t = 4t_0$

Open solution

Q130. AIEEE 2004

A particle at the end of a spring executes $S.H.M$ with a period t_1. While the corresponding period for another spring is t_2. If the period of oscillation with the two springs in series is T then

(A) $T^{-1} = t_1^{-1} + t_2^{-1}$ (B) $T^2 = t_1^2 + t_2^2$

(C) $T = t_1 + t_2$ (D) $T^{-2} = t_1^{-2} + t_2^{-2}$

Open solution

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Q131. AIEEE 2004

The total energy of particle, executing simple harmonic motion is

(A) independent of x (B) $\propto x^2$ (C) $\propto x$ (D) $\propto x^{1/2}$

Open solution

Q132. AIEEE 2004

A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0. An external force $F(t)$ proportional to $\cos \omega t$ $(\omega \neq \omega_0)$ is applied to the oscillator. The time displacement of the oscillator will be proportional to

(A) $\frac{1}{m(\omega_0^2+\omega^2)}$ (B) $\frac{1}{m(\omega_0^2-\omega^2)}$

(C) $\frac{m}{\omega_0^2-\omega^2}$ (D) $\frac{m}{\omega_0^2+\omega^2}$

Open solution

Q133. AIEEE 2003

A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes SHM of time period T. If the mass is increased by m. the time period becomes $\frac{5T}{3}$. Then the ratio of $\frac{m}{M}$ is

(A) $\frac{3}{5}$ (B) $\frac{25}{9}$ (C) $\frac{16}{9}$ (D) $\frac{5}{3}$

Open solution

Q134. AIEEE 2003

Two particles A and B of equal masses are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the maximum velocities, during oscillation, are equal, the ratio of amplitude of A and B is

(A) $\sqrt{\frac{k_1}{k_2}}$ (B) $\frac{k_2}{k_1}$ (C) $\sqrt{\frac{k_2}{k_1}}$ (D) $\frac{k_1}{k_2}$

Open solution

Q135. AIEEE 2003

The length of a simple pendulum executing simple harmonic motion is increased by 21%. The percentage increase in the time period of the pendulum of increased length is

(A) 11% (B) 21% (C) 42% (D) 10%

Open solution

Q136. AIEEE 2003

The displacement of particle varies according to the relation $x = 4(\cos \pi t + \sin \pi t)$. The amplitude of the particle is

(A) -4 (B) 4 (C) $4\sqrt{2}$ (D) 8

Open solution

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Q137. AIEEE 2003

A body executes simple harmonic motion. The potential energy $(P.E)$, the kinetic energy $(K.E)$ and total energy $(T.E)$ are measured as a function of displacement x. Which of the following statements is true ?

(A) $K.E$ is maximum when $x = 0$

(B) $T.E$ is zero when $x = 0$

(C) $K.E$ is maximum when x is maximum

(D) $P.E$ is maximum when $x = 0$

Open solution

Q138. AIEEE 2002

In a simple harmonic oscillator, at the mean position

(A) kinetic energy is minimum, potential energy is maximum

(B) both kinetic and potential energies are maximum

(C) kinetic energy is maximum, potential energy is minimum

(D) both kinetic and potential energies are minimum.

Open solution

Q139. AIEEE 2002

A child swinging on a swing in sitting position, stands up, then the time period of the swing will

(A) increase

(B) decrease

(C) remains same

(D) increases of the child is long and decreases if the child is short

Open solution

Q140. AIEEE 2002

If a spring has time period T, and is cut into n equal parts, then the time period of each part will be

(A) $T\sqrt{n}$ (B) $T/\sqrt{n}$ (C) nT (D) T

Open solution

Numerical 39 questions

Q141. JEE Main 2026 (Online) 4th April Morning Shift

The velocity of a particle executing simple harmonic motion along x-axis is described as $v^2 = 50 - x^2$, where x represents displacement. If the time period of motion is $\frac{x}{7}$ s, the value of x is ____ .

Open solution

Q142. JEE Main 2026 (Online) 28th January Morning Shift

The displacement of a particle, executing simple harmonic motion with time period T, is expressed as $x(t) = A\sin\omega t$, where A is the amplitude. The maximum value of potential energy of this oscillator is found at $t = T/2\beta$. The value of β is ____ .

Open solution

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Q143. JEE Main 2024 (Online) 9th April Evening Shift

A particle of mass $0.50\ \text{kg}$ executes simple harmonic motion under force $F = -50(\text{Nm}^{-1})x$. The time period of oscillation is $\frac{x}{35}s$. The value of x is _____.

(Given $\pi = \frac{22}{7}$)

Open solution

Q144. JEE Main 2024 (Online) 9th April Morning Shift

The position, velocity and acceleration of a particle executing simple harmonic motion are found to have magnitudes of $4\ \text{m}, 2\ \text{ms}^{-1}$ and $16\ \text{ms}^{-2}$ at a certain instant. The amplitude of the motion is $\sqrt{x}\ \text{m}$ where x is _____.

Open solution

Q145. JEE Main 2024 (Online) 8th April Evening Shift

An object of mass $0.2\ \text{kg}$ executes simple harmonic motion along x axis with frequency of $\left(\frac{25}{\pi}\right)\text{Hz}$. At the position $x = 0.04\ \text{m}$ the object has kinetic energy $0.5\ \text{J}$ and potential energy $0.4\ \text{J}$. The amplitude of oscillation is _____ cm.

Open solution

Q146. JEE Main 2024 (Online) 6th April Morning Shift

A particle is doing simple harmonic motion of amplitude $0.06\ \text{m}$ and time period $3.14\ \text{s}$. The maximum velocity of the particle is _____ cm/s.

Open solution

Q147. JEE Main 2024 (Online) 4th April Evening Shift

The displacement of a particle executing SHM is given by $x = 10\sin\left(wt + \frac{\pi}{3}\right)m$. The time period of motion is $3.14\ \text{s}$. The velocity of the particle at $t = 0$ is _____ m/s.

Open solution

Q148. JEE Main 2024 (Online) 1st February Evening Shift

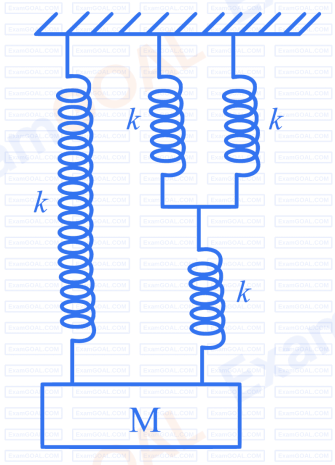
A mass m is suspended from a spring of negligible mass and the system oscillates with a frequency f_1. The frequency of oscillations if a mass $9\ \text{m}$ is suspended from the same spring is f_2. The value of $\frac{f_1}{f_2}$ is _____.

Open solution

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Q149. JEE Main 2024 (Online) 31st January Evening Shift

The time period of simple harmonic motion of mass M in the given figure is $\pi\sqrt{\frac{\alpha M}{5k}}$, where the value of α is _____.



Open solution

Q150. JEE Main 2024 (Online) 31st January Morning Shift

A particle performs simple harmonic motion with amplitude A. Its speed is increased to three times at an instant when its displacement is $\frac{2A}{3}$. The new amplitude of motion is $\frac{nA}{3}$. The value of n is _____.

Open solution

Q151. JEE Main 2024 (Online) 29th January Evening Shift

A simple harmonic oscillator has an amplitude A and time period 6π second. Assuming the oscillation starts from its mean position, the time required by it to travel from $x = \text{A}$ to $x = \frac{\sqrt{3}}{2}$ A will be $\frac{\pi}{x}$ s, where $x =$ _____.

Open solution

Q152. JEE Main 2024 (Online) 29th January Morning Shift

When the displacement of a simple harmonic oscillator is one third of its amplitude, the ratio of total energy to the kinetic energy is $\frac{x}{8}$, where $x =$ _____.

Open solution

Q153. JEE Main 2024 (Online) 27th January Morning Shift

A particle executes simple harmonic motion with an amplitude of $4\ \text{cm}$. At the mean position, velocity of the particle is $10\ \text{cm/s}$. The distance of the particle from the mean position when its speed becomes $5\ \text{cm/s}$ is $\sqrt{\alpha}\ \text{cm}$, where $\alpha =$ _____.

Open solution

Q154. JEE Main 2023 (Online) 13th April Morning Shift

At a given point of time the value of displacement of a simple harmonic oscillator is given as $\text{y} = \text{A}\cos(30^\circ)$. If amplitude is $40\ \text{cm}$ and kinetic energy at that time is $200\ \text{J}$, the value of force constant is $1.0 \times 10^x\ \text{Nm}^{-1}$. The value of x is _____.

Open solution

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Q155. JEE Main 2023 (Online) 10th April Evening Shift

A rectangular block of mass $5\ \mathrm{kg}$ attached to a horizontal spiral spring executes simple harmonic motion of amplitude $1\ \mathrm{m}$ and time period $3.14\ \mathrm{s}$. The maximum force exerted by spring on block is _____ N

Open solution

Q156. JEE Main 2023 (Online) 6th April Evening Shift

A simple pendulum with length $100\ \mathrm{cm}$ and bob of mass $250\ \mathrm{g}$ is executing S.H.M. of amplitude $10\ \mathrm{cm}$. The maximum tension in the string is found to be $\frac{x}{40}\ \mathrm{N}$. The value of x is _____.

Open solution

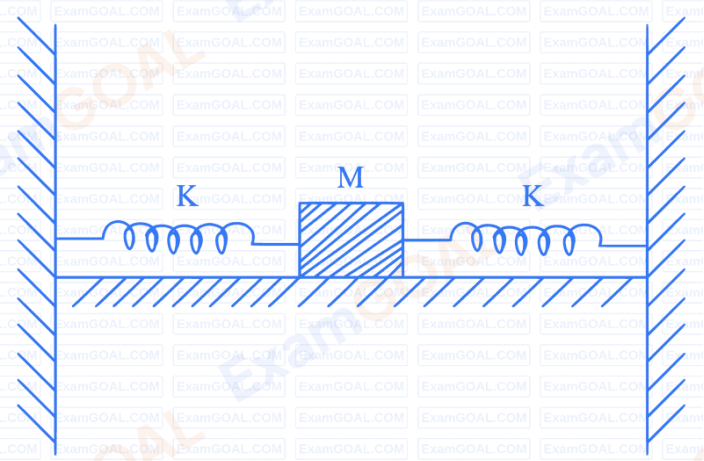
Q157. JEE Main 2023 (Online) 1st February Morning Shift

The amplitude of a particle executing SHM is $3\ \mathrm{cm}$. The displacement at which its kinetic energy will be 25% more than the potential energy is: _____ cm

Open solution

Q158. JEE Main 2023 (Online) 31st January Morning Shift

In the figure given below, a block of mass $M = 490\ \mathrm{g}$ placed on a frictionless table is connected with two springs having same spring constant $\left(\mathrm{K} = 2\ \mathrm{N\ m^{-1}}\right)$. If the block is horizontally displaced through 'X' m then the number of complete oscillations it will make in 14π seconds will be _____.



Open solution

Q159. JEE Main 2023 (Online) 30th January Evening Shift

The velocity of a particle executing SHM varies with displacement (x) as $4v^2 = 50 - x^2$. The time period of oscillations is $\frac{x}{7}s$. The value of x is _____. (Take $\pi = \frac{22}{7}$)

Open solution

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Q160. JEE Main 2023 (Online) 30th January Morning Shift

The general displacement of a simple harmonic oscillator is $x = A\sin\omega t$. Let T be its time period. The slope of its potential energy (U) - time (t) curve will be maximum when $t = \frac{T}{\beta}$. The value of β is _____.

Open solution

Q161. JEE Main 2023 (Online) 29th January Evening Shift

A particle of mass 250 g executes a simple harmonic motion under a periodic force $\mathrm{F} = (-25\,x)\mathrm{N}$. The particle attains a maximum speed of 4 m/s during its oscillation. The amplitude of the motion is _____ cm.

Open solution

Q162. JEE Main 2023 (Online) 24th January Evening Shift

A mass m attached to free end of a spring executes SHM with a period of 1s. If the mass is increased by 3 kg the period of the oscillation increases by one second, the value of mass m is _____ kg.

Open solution

Q163. JEE Main 2023 (Online) 24th January Morning Shift



A block of mass 2 kg is attached with two identical springs of spring constant 20 N/m each. The block is placed on a frictionless surface and the ends of the springs are attached to rigid supports (see figure). When the mass is displaced from its equilibrium position, it executes a simple harmonic motion. The time period of oscillation is $\frac{\pi}{\sqrt{x}}$ in SI unit. The value of x is _____.

Open solution

Q164. JEE Main 2022 (Online) 30th July Evening Shift

A particle executing SHM has its velocity $20\ \mathrm{cm\,s^{-1}}$ at mean position and acceleration of $25\ \mathrm{cm\,s^{-2}}$ at one of its extreme position. The amplitude of the particle will be ____ cm.

Open solution

Q165. JEE Main 2022 (Online) 29th July Evening Shift

The metallic bob of simple pendulum has the relative density 5. The time period of this pendulum is $10\ \mathrm{s}$. If the metallic bob is immersed in water, then the new time period becomes $5\sqrt{x}$ s. The value of x will be _____.

Open solution

Q166. JEE Main 2022 (Online) 28th July Evening Shift

The potential energy of a particle of mass $4\ \mathrm{kg}$ in motion along the x-axis is given by $\mathrm{U} = 4(1-\cos 4x)$ J. The time period of the particle for small oscillation $(\sin\theta \simeq \theta)$ is $\left(\frac{\pi}{K}\right)s$. The value of K is _____.

Open solution

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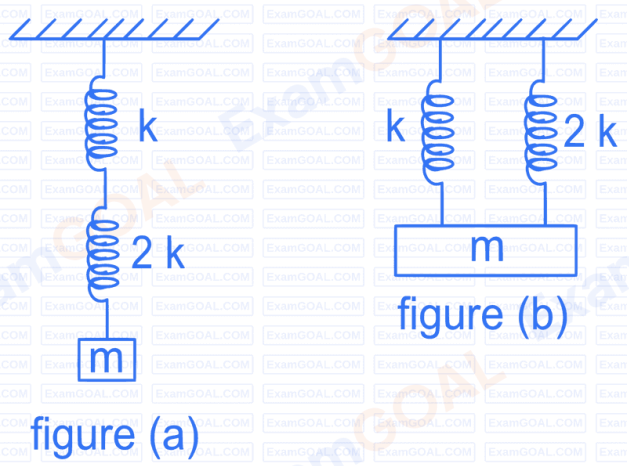
Q167. JEE Main 2022 (Online) 27th July Morning Shift

A mass $0.9\ \mathrm{kg}$, attached to a horizontal spring, executes SHM with an amplitude A_1. When this mass passes through its mean position, then a smaller mass of $124\ \mathrm{g}$ is placed over it and both masses move together with amplitude A_2. If the ratio $\frac{A_1}{A_2}$ is $\frac{\alpha}{\alpha-1}$, then the value of α will be _____.

Open solution

Q168. JEE Main 2022 (Online) 26th July Evening Shift

As per given figures, two springs of spring constants k and $2k$ are connected to mass m. If the period of oscillation in figure (a) is $3\mathrm{s}$, then the period of oscillation in figure (b) will be $\sqrt{x}\ s$. The value of x is _____.



Open solution

Q169. JEE Main 2022 (Online) 29th June Morning Shift

A body is performing simple harmonic with an amplitude of 10 cm. The velocity of the body was tripled by air jet when it is at 5 cm from its mean position. The new amplitude of vibration is $\sqrt{x}$ cm. The value of x is _____.

Open solution

Q170. JEE Main 2022 (Online) 27th June Evening Shift

A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is 6 s. The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is _____ s.

Open solution

Q171. JEE Main 2021 (Online) 31st August Morning Shift

A particle of mass 1 kg is hanging from a spring of force constant 100 Nm^{-1}. The mass is pulled slightly downward and released so that it executes free simple harmonic motion with time period T. The time when the kinetic energy and potential energy of the system will become equal, is $\frac{T}{x}$. The value of x is _____.

Open solution

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Q172. JEE Main 2021 (Online) 27th August Evening Shift

Two simple harmonic motion, are represented by the equations $y_1 = 10\sin\left(3\pi t + \frac{\pi}{3}\right)$ $y_2 = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)$ Ratio of amplitude of y_1 to y_2 = x : 1. The value of x is _____.

Open solution

Q173. JEE Main 2021 (Online) 26th August Evening Shift

Two simple harmonic motions are represented by the equations

$x_1 = 5\sin\left(2\pi t + \frac{\pi}{4}\right)$ and $x_2 = 5\sqrt{2}(\sin 2\pi t + \cos 2\pi t)$. The amplitude of second motion is ..... times the amplitude in first motion.

Open solution

Q174. JEE Main 2021 (Online) 27th July Evening Shift

A particle executes simple harmonic motion represented by displacement function as

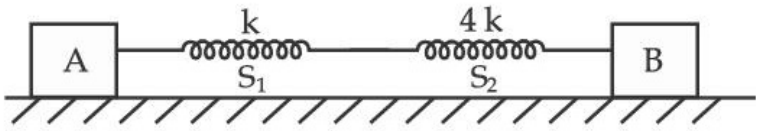
x(t) = A sin(ωt + ϕ)

If the position and velocity of the particle at t = 0 s are 2 cm and 2ω cm s^{-1} respectively, then its amplitude is $x\sqrt{2}$ cm where the value of x is _____.

Open solution

Q175. JEE Main 2021 (Online) 25th July Morning Shift

In the reported figure, two bodies A and B of masses 200 g and 800 g are attached with the system of springs. Springs are kept in a stretched position with some extension when the system is released. The horizontal surface is assumed to be frictionless. The angular frequency will be _____ rad/s when k = 20 N/m.



Open solution

Q176. JEE Main 2021 (Online) 18th March Morning Shift

A particle performs simple harmonic motion with a period of 2 second. The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}$s. The value of 'a' to the nearest integer is _____.

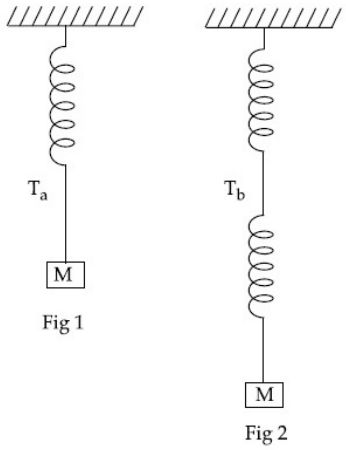
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Q177. JEE Main 2021 (Online) 17th March Morning Shift

Consider two identical springs each of spring constant k and negligible mass compared to the mass M as shown. Fig. 1 shows one of them and Fig. 2 shows their series combination. The ratios of time period of oscillation of the two SHM is $T_b/T_a = \sqrt{x}$, where value of x is _____. (Round off to the Nearest Integer)

Open solution



Q178. JEE Main 2021 (Online) 26th February Evening Shift

A particle executes S.H.M. with amplitude 'a', and time period 'T'. The displacement of the particle when its speed is half of maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is _____.

Open solution

Q179. JEE Main 2021 (Online) 26th February Evening Shift

Time period of a simple pendulum is T. The time taken to complete $\frac{5}{8}$ oscillations starting from mean position is $\frac{\alpha}{\beta}T$. The value of α is _____.

Open solution

MCQ (More than One Correct Answer) 2 questions

Q180. AIEEE 2009

If x, v and a denote the displacement, the velocity and the acceleration of a particle executing simple harmonic motion of time period T, then, which of the following does not change with time?

(A) aT/x (B) $aT + 2\pi v$ (C) aT/v (D) $a^2T^2 + 4\pi^2v^2$

Open solution

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Q181. AIEEE 2006

A coin is placed on a horizontal platform which undergoes vertical simple harmonic motion of angular frequency ω. The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time

Open solution

(A) at the mean position of the platform

(B) for an amplitude of $\frac{g}{\omega^2}$

(C) for an amplitude of $\frac{g^2}{\omega^2}$

(D) at the highest position of the platform

Out of syllabus 13 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q182. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

A block of mass 1 kg attached to a spring is made to oscillate with an initial amplitude of 12 cm. After 2 minutes the amplitude decreases to 6 cm. Determine the value of the damping constant for this motion . (take ln 2 = 0.693)

Open solution

(A) 0.69×10^2 kg s^{-1}

(B) 3.3×10^2 kg s^{-1}

(C) 1.16×10^{-2} kg s^{-1}

(D) 5.7×10^{-3} kg s^{-1}

Q183. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

Amplitude of a mass-spring system, which is executing simple harmonic motion decreases with time. If mass = 500g, Decay constant = 20 g/s then how much time is required for the amplitude of the system to drop to half of its initial value? (ln 2 = 0.693)

Open solution

(A) 17.32 s (B) 34.65 s (C) 0.034 s (D) 15.01 s

Q184. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The displacement of a damped harmonic oscillator is given by

$x(t) = e^{-0.1t} \cos(10\pi t + f)$.

Here t is in seconds. The time taken for its amplitude of vibration to drop to half of its initial value is close to :

Open solution

(A) 27 s (B) 13 s (C) 7 s (D) 4 s

Q185. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to 1/1000 of the original amplitude is close to :-

Open solution

(A) 100 s (B) 10 s (C) 20 s (D) 50 s

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Q186. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is best given by :

Open solution

(A) 1 rad/s (B) 10^{-3} rad/s (C) 10^{-1} rad/s (D) 10^{-5} rad/s

Q187. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

A block of mass 0.1 kg is connected to an elastic spring of spring constant 640 Nm^{-1} and oscillates in a damping medium of damping constant 10^{-2} kg s^{-1} . The system dissipates its energy gradually. The time taken for its mechanical energy of vibration to drop to half of its initial value, is closest to :

Open solution

(A) 2 s (B) 3.5 s (C) 5 s (D) 7 s

Q188. JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

A pendulum with time period of 1s is losing energy due to damping. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s^{-1}) is :

Open solution

(A) $\frac{1}{30}\ln 3$ (B) $\frac{1}{15}\ln 3$ (C) 2 (D) $\frac{1}{2}$

Q189. JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

A simple harmonic oscillator of angular frequency 2 rad s^{-1} is acted upon by an external force F = sint N. If the oscillator is at rest in its equilibrium position at t = 0, its position at later times is proportional to :

Open solution

(A) sint + $\frac{1}{2}$ sin 2t (B) sint + $\frac{1}{2}$ cos 2t
(C) cost − $\frac{1}{2}$ sin 2t (D) sint − $\frac{1}{2}$ sin 2t

Q190. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

The angular frequency of the damped oscillator is given by,

Open solution

$\omega = \sqrt{\left(\frac{k}{m} - \frac{r^2}{4m^2}\right)}$ where k is the spring constant, m is the mass of the oscillator and r is the damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period compared to the undamped oscillator is approximately as follows :

(A) increases by 1% (B) increases by 8% (C) decreases 1% (D) decreases 8%

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Q191. JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10 cm to 8 cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3, the time in which amplitude of this pendulum will reduce from 10 cm to 5 cm in carbon dioxide will be close to (ln 5 = 1.609, ln 2 = 0.693).

Open solution

(A) 231 s (B) 208 s (C) 161 s (D) 142 s

Q192. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in $5s$. In another $10s$ it will decrease to α times its original magnitude, where α equals

Open solution

(A) 0.7 (B) 0.81 (C) 0.729 (D) 0.6

Q193. AIEEE 2012 · MCQ (Single Correct Answer)

If a simple pendulum has significant amplitude (up to a factor of $1/e$ of original) only in the period between $t = 0s$ to $t = \tau\, s$, then τ may be called the average life of the pendulum When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity with b as the constant of proportionality, the average life time of the pendulum is (assuming damping is small) in seconds :

Open solution

(A) $\frac{0.693}{b}$ (B) b (C) $\frac{1}{b}$ (D) $\frac{2}{b}$

Q194. AIEEE 2004 · MCQ (Single Correct Answer)

In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force while the energy is maximum for a frequency ω_2 of the force; then

Open solution

(A) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large

(B) $\omega_1 > \omega_2$

(C) $\omega_1 = \omega_2$

(D) $\omega_1 < \omega_2$

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **C**	4. **A**	5. **A**
6. **A**	7. **C**	8. **C**	9. **C**	10. **B**
11. **A**	12. **B**	13. **A**	14. **C**	15. **D**
16. **B**	17. **C**	18. **D**	19. **D**	20. **C**
21. **A**	22. **D**	23. **A**	24. **D**	25. **C**
26. **A**	27. **A**	28. **D**	29. **C**	30. **B**
31. **D**	32. **C**	33. **A**	34. **D**	35. **C**
36. **A**	37. **A**	38. **C**	39. **B**	40. **A**
41. **C**	42. **B**	43. **A**	44. **C**	45. **A**
46. **B**	47. **D**	48. **C**	49. **B**	50. **C**
51. **D**	52. **A**	53. **D**	54. **A**	55. **D**
56. **B**	57. **B**	58. **C**	59. **D**	60. **D**
61. **D**	62. **C**	63. **B**	64. **C**	65. **B**
66. **A**	67. **C**	68. **A**	69. **B**	70. **A**
71. **B**	72. **C**	73. **D**	74. **D**	75. **D**
76. **B**	77. **D**	78. **B**	79. **C**	80. **D**
81. **B**	82. **B**	83. **D**	84. **B**	85. **D**
86. **D**	87. **C**	88. **C**	89. **A**	90. **A**
91. **C**	92. **C**	93. **B**	94. **C**	95. **B**
96. **A**	97. **D**	98. **A**	99. **B**	100. **C**
101. **C**	102. **C**	103. **D**	104. **C**	105. **B**
106. **D**	107. **C**	108. **A**	109. **D**	110. **D**
111. **D**	112. **A**	113. **A**	114. **C**	115. **A**
116. **A**	117. **A**	118. **C**	119. **A**	120. **B**
121. **B**	122. **A**	123. **A**	124. **A**	125. **B**
126. **C**	127. **B**	128. **A**	129. **A**	130. **B**
131. **A**	132. **B**	133. **C**	134. **C**	135. **D**
136. **C**	137. **A**	138. **C**	139. **B**	140. **B**

Numerical

141. **44**	142. **2**	143. **22**	144. **17**	145. **6**

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146. **12**	147. **10**	148. **3**	149. **12**	150. **7**
151. **2**	152. **9**	153. **12**	154. **4**	155. **20**
156. **101**	157. **2**	158. **20**	159. **88**	160. **8**
161. **40**	162. **1**	163. **5**	164. **16**	165. **5**
166. **2**	167. **16**	168. **2**	169. **700**	170. **1**
171. **8**	172. **1**	173. **2**	174. **2**	175. **10**
176. **6**	177. **2**	178. **3**	179. **7**	

MCQ (More than One Correct Answer)

180. **A, D**	181. **B, D**

Out of syllabus

182. **C**	183. **B**	184. **C**	185. **C**	186. **B**
187. **D**	188. **A**	189. **D**	190. **A**	191. **C**
192. **C**	193. **D**	194. **C**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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